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Bashir et al.

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(54) **SYSTEM AND METHOD TO MAP HIERARCHICAL MULTI-TENANT ACCESS TO SERVICES**

(71) Applicant: **DISH Wireless L.L.C.**, Englewood, CO (US)

(72) Inventors: **Kazi Bashir**, Lewisville, TX (US); **Rajesh Chilka**, Littleton, CO (US); **Mehdi Alasti**, Arlington, VA (US); **Siddhartha Chenumolu**, Ashburn, VA (US)

(73) Assignee: **DISH Wireless L.L.C.**, Englewood, CO (US)

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H04L 9/40 (2022.01)

(52) **U.S. Cl.**
CPC ***H04L 63/102*** (2013.01); ***H04L 63/105*** (2013.01); ***H04L 63/20*** (2013.01)

(58) **Field of Classification Search**
CPC H04L 63/102; H04L 63/105; H04L 63/20
See application file for complete search history.

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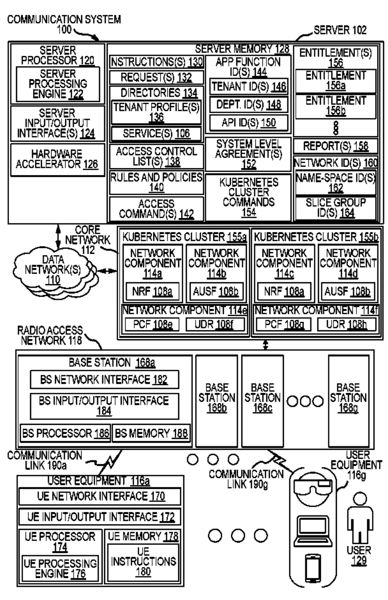
Primary Examiner — El Hadji M Sall
(74) *Attorney, Agent, or Firm* — Baker Botts L.L.P.

(57) **ABSTRACT**

An apparatus comprises a memory and a processor communicatively coupled to one another. The memory may be configured to store one or more directories comprising access to multiple tenant profiles and one or more network access commands configured to provide access to one or more entitlements. Each tenant profile of the tenant profiles are associated with one or more services. The processor may be configured to receive a request to access at least one service. The request comprises an application function identifier (AFID) comprising a tenant ID that references a tenant profile of the tenant profiles, a department ID that references multiple entitlements associated with the tenant profile, and an API ID that references a service associated with the entitlements. Further, the processor may be configured to generate a report comprising multiple network access commands configured to enable access to the service in accordance with the entitlements.

20 Claims, 9 Drawing Sheets



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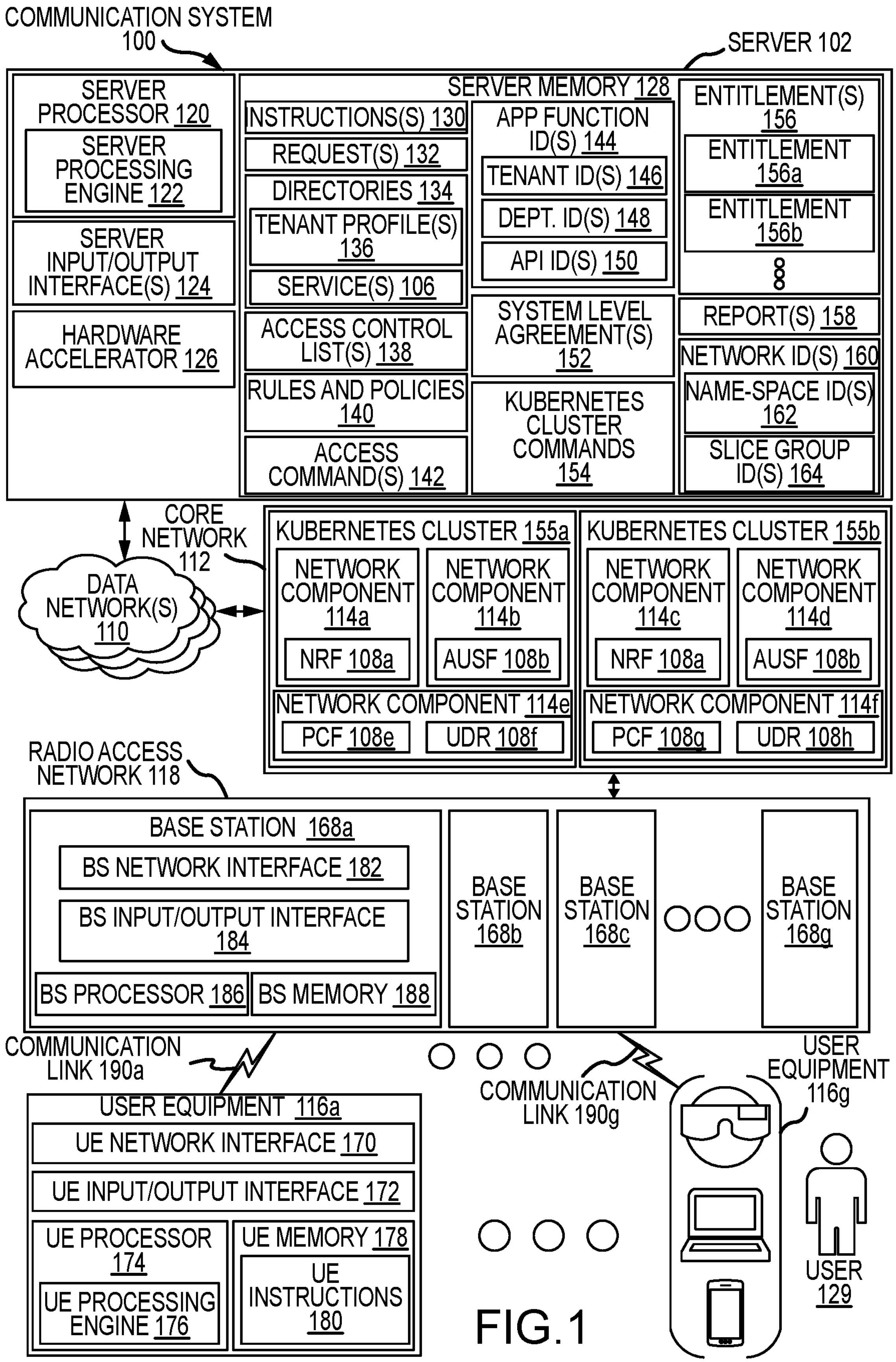
COMMUNICATION SYSTEM 100
SERVER 102
SERVER PROCESSOR 120
SERVER PROCESSING ENGINE 122
SERVER INPUT/OUTPUT INTERFACE(S) 124
HARDWARE ACCELERATOR 126
SERVER MEMORY 128
INSTRUCTIONS(S) 130
REQUEST(S) 132
DIRECTORIES 134
TENANT PROFILE(S) 136
SERVICE(S) 106
ACCESS CONTROL LIST(S) 138
RULES AND POLICIES 140
ACCESS COMMAND(S) 142
APP FUNCTION ID(S) 144
TENANT ID(S) 146
DEPT. ID(S) 148
API ID(S) 150
SYSTEM LEVEL AGREEMENT(S) 152
KUBERNETES CLUSTER COMMANDS 154
ENTITLEMENT(S) 156
ENTITLEMENT 156a
ENTITLEMENT 156b
REPORT(S) 158
NETWORK ID(S) 160
NAME-SPACE ID(S) 162
SLICE GROUP ID(S) 164
CORE NETWORK 112
DATA NETWORK(S) 110
KUBERNETES CLUSTER 155a
KUBERNETES CLUSTER 155b
NETWORK COMPONENT 114a
NRF 108a
NETWORK COMPONENT 114b
AUSF 108b
NETWORK COMPONENT 114c
NRF 108a
NETWORK COMPONENT 114d
AUSF 108b
NETWORK COMPONENT 114e
PCF 108e
UDR 108f
NETWORK COMPONENT 114f
PCF 108g
UDR 108h
RADIO ACCESS NETWORK 118
BASE STATION 168a
BS NETWORK INTERFACE 182
BS INPUT/OUTPUT INTERFACE 184
BS PROCESSOR 186
BS MEMORY 188
BASE STATION 168b
BASE STATION 168c
BASE STATION 168g
COMMUNICATION LINK 190a
USER EQUIPMENT 116a
UE NETWORK INTERFACE 170
UE INPUT/OUTPUT INTERFACE 172
UE PROCESSOR 174
UE PROCESSING ENGINE 176
UE MEMORY 178
UE INSTRUCTIONS 180
COMMUNICATION LINK 190g
USER EQUIPMENT 116g
USER 129

FIG.1

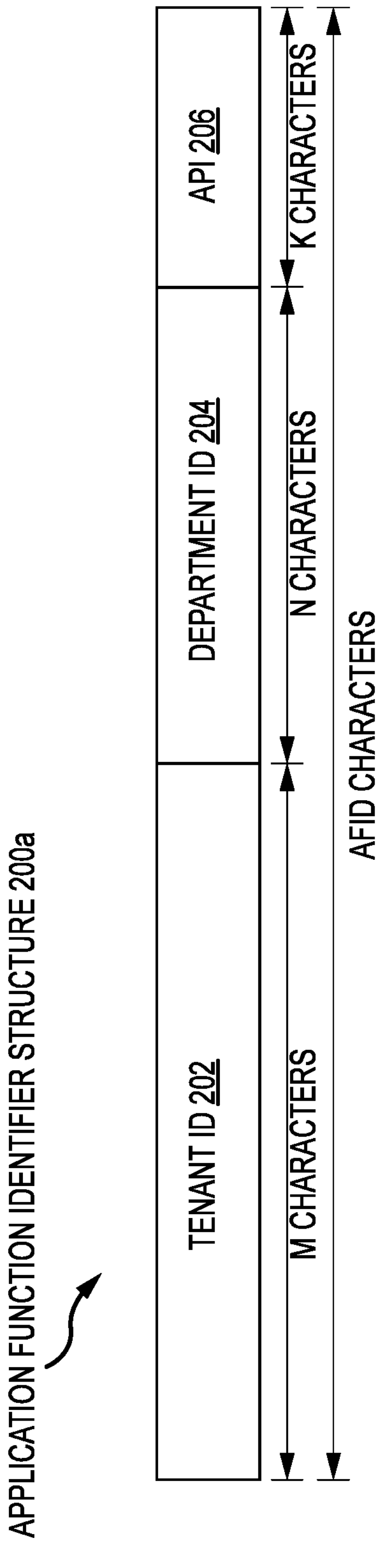
APPLICATION FUNCTION IDENTIFIER STRUCTURE 200a
TENANT ID 202
DEPARTMENT ID 204
API 206
M CHARACTERS
N CHARACTERS
K CHARACTERS
AFID CHARACTERS

FIG 2A

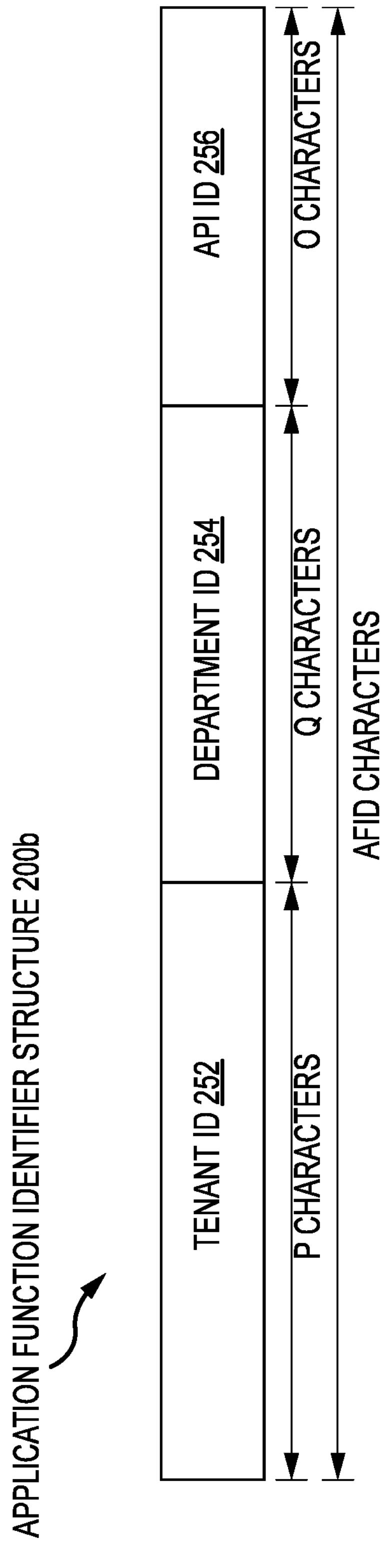
APPLICATION FUNCTION IDENTIFIER STRUCTURE 200b
TENANT ID 252
DEPARTMENT ID 254
API ID 256
P CHARACTERS
Q CHARACTERS
O CHARACTERS
AFID CHARACTERS

FIG.2B

APPLICATION FUNCTION IDENTIFIER MAPPING OPERATIONS 300

ENTITLEMENT 156a ENTITLEMENT 156b ENTITLEMENT 156c ENTITLEMENT 156d

	AFID			ACCESS TO MONITORING INFORMATION	EVENT CONTROLS	DIRECTORY MANAGEMENT	SYSTEM ACCESS
	TENANT-ID	DEPARTMENT-ID	API-ID				
AFID 310	TENANT1	DEPARTMENT1	APPLICATION1	x			
AFID 312	TENANT1	DEPARTMENT2	APPLICATION2		x	x	
AFID 314	TENANT1	DEPARTMENT2	APPLICATION3	x			x
AFID 316	TENANT1	DEPARTMENT3	APPLICATION3	x			x
AFID 320	TENANT2	DEPARTMENT1	APPLICATION1	x			

FIG.3

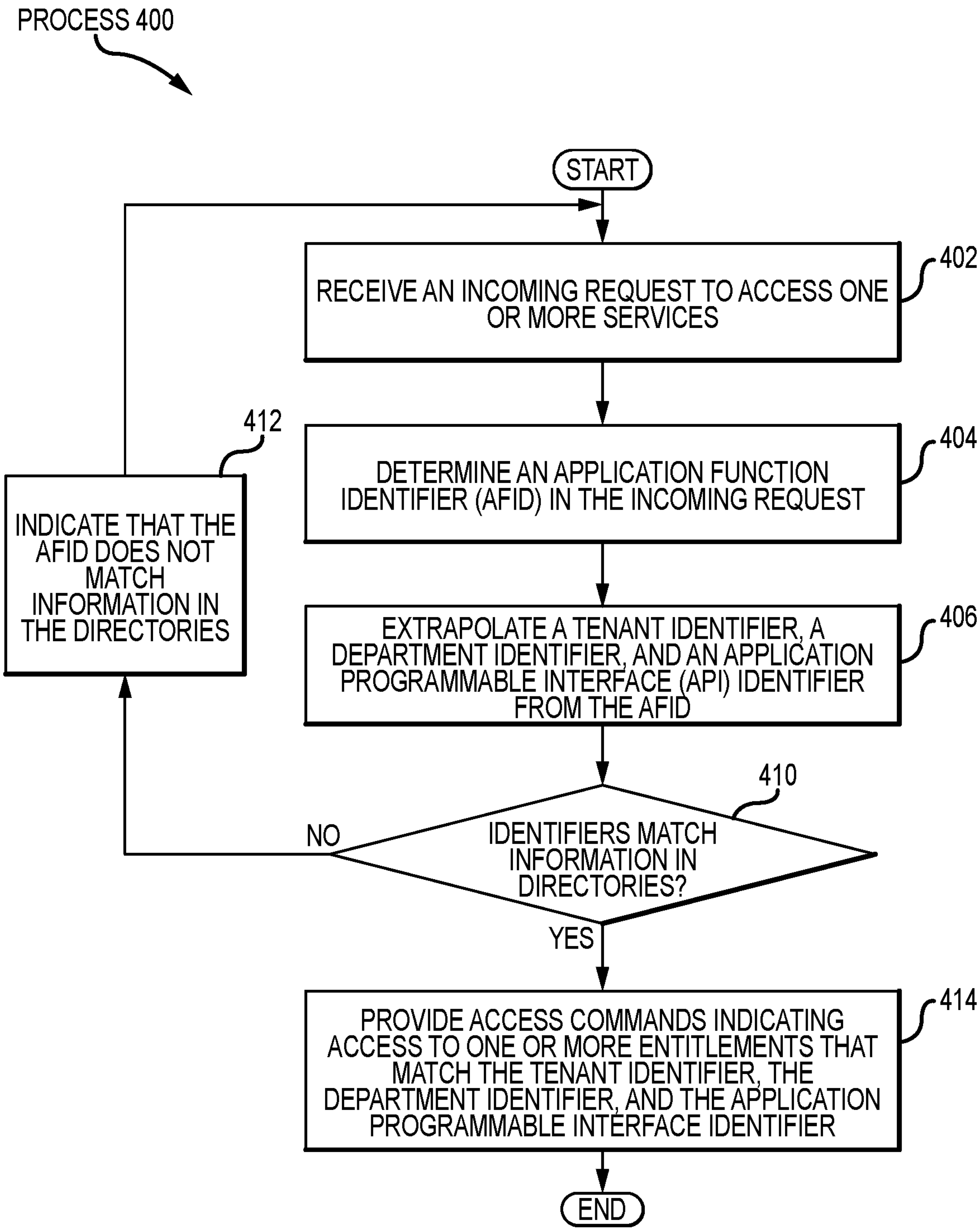
PROCESS 400
START
RECEIVE AN INCOMING REQUEST TO ACCESS ONE OR MORE SERVICES
402
DETERMINE AN APPLICATION FUNCTION IDENTIFIER (AFID) IN THE INCOMING REQUEST
404
EXTRAPOLATE A TENANT IDENTIFIER, A DEPARTMENT IDENTIFIER, AND AN APPLICATION PROGRAMMABLE INTERFACE (API) IDENTIFIER FROM THE AFID
406
IDENTIFIERS MATCH INFORMATION IN DIRECTORIES?
410
NO
YES
INDICATE THAT THE AFID DOES NOT MATCH INFORMATION IN THE DIRECTORIES
412
PROVIDE ACCESS COMMANDS INDICATING ACCESS TO ONE OR MORE ENTITLEMENTS THAT MATCH THE TENANT IDENTIFIER, THE DEPARTMENT IDENTIFIER, AND THE APPLICATION PROGRAMMABLE INTERFACE IDENTIFIER
414
END

FIG.4

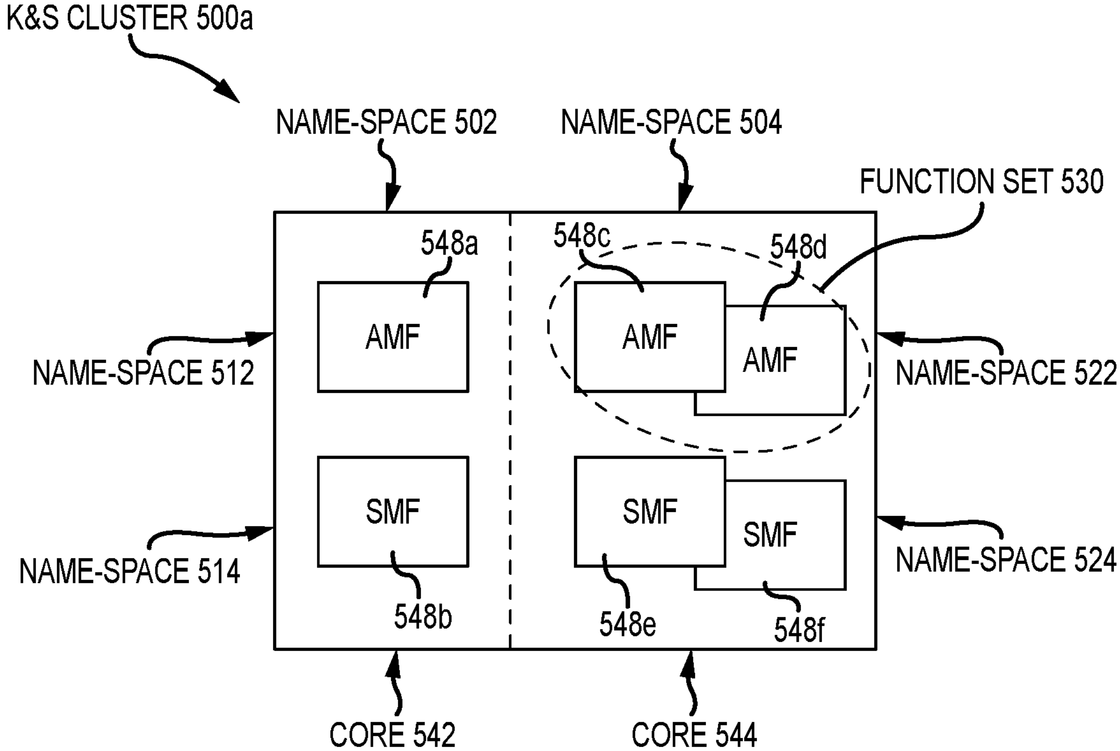
K&S CLUSTER 500a
NAME-SPACE 502
NAME-SPACE 504
FUNCTION SET 530
548a
548c
548d
AMF
AMF
AMF
NAME-SPACE 512
NAME-SPACE 522
SMF
SMF
SMF
NAME-SPACE 514
NAME-SPACE 524
548b
548e
548f
CORE 542
CORE 544

FIG.5A

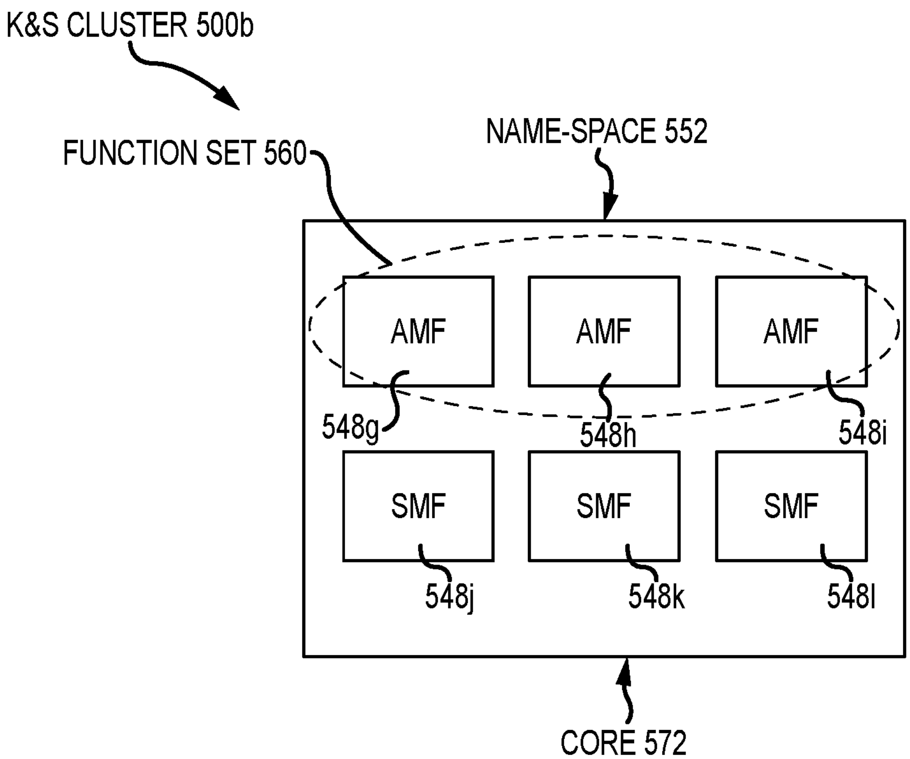
K&S CLUSTER 500b
FUNCTION SET 560
NAME-SPACE 552
AMF
AMF
AMF
548g
548h
548i
SMF
SMF
SMF
548j
548k
548l
CORE 572

FIG.5B

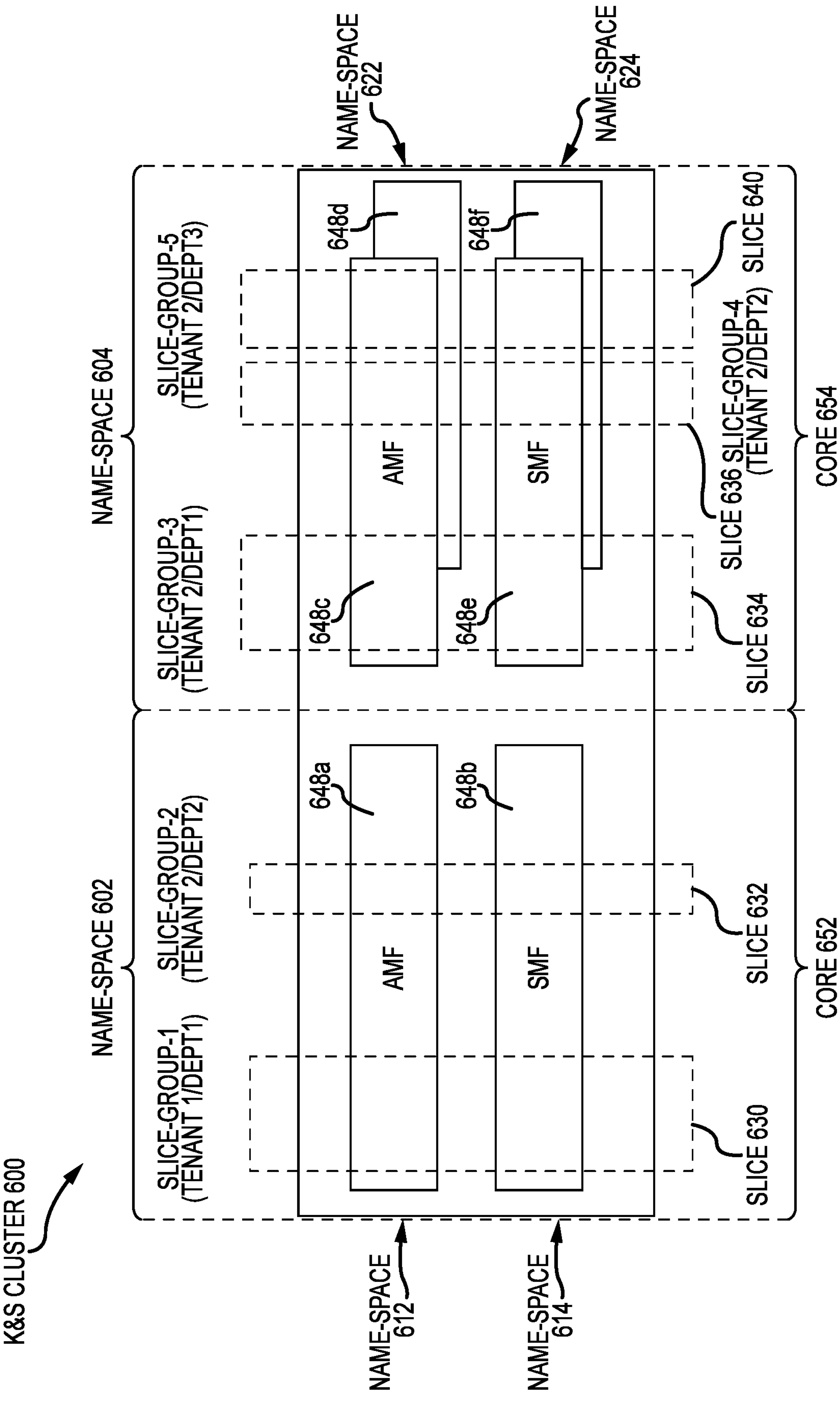
K&S CLUSTER 600
NAME-SPACE 602
NAME-SPACE 604
SLICE-GROUP-1 (TENANT 1/DEPT1)
SLICE-GROUP-2 (TENANT 2/DEPT2)
SLICE-GROUP-3 (TENANT 2/DEPT1)
SLICE-GROUP-5 (TENANT 2/DEPT3)
SLICE-GROUP-4 (TENANT 2/DEPT2)
NAME-SPACE 612
NAME-SPACE 614
NAME-SPACE 622
NAME-SPACE 624
AMF
SMF
AMF
SMF
648a
648b
648c
648d
648e
648f
SLICE 630
SLICE 632
SLICE 634
SLICE 636
SLICE 640
CORE 652
CORE 654

FIG.6

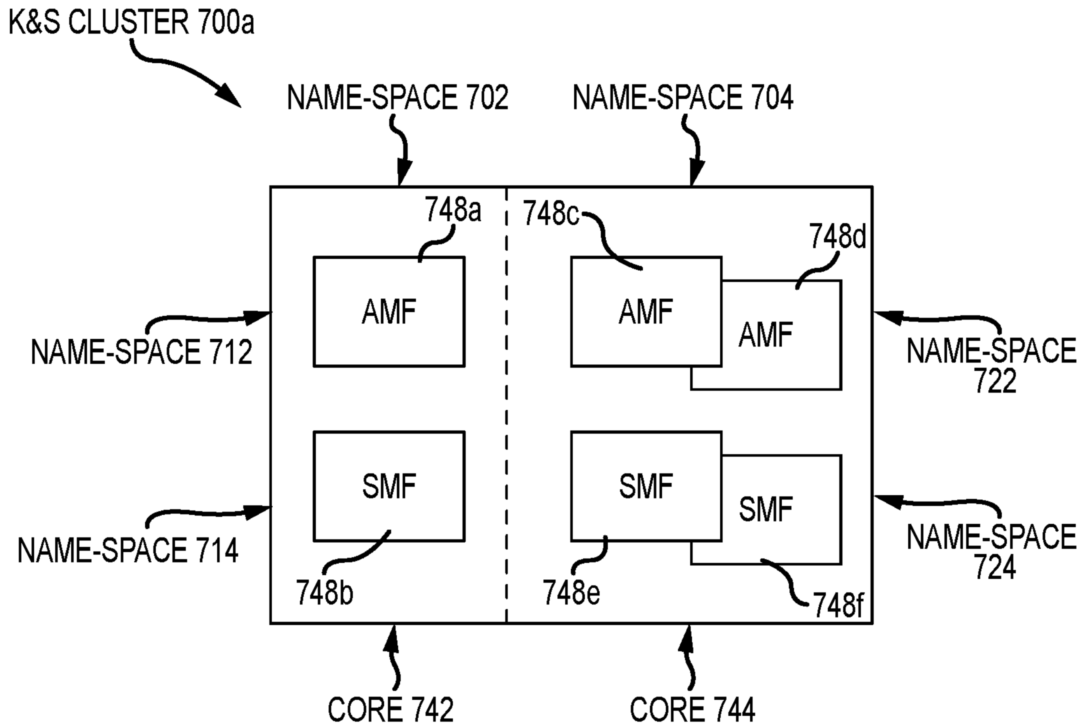
K&S CLUSTER 700a
NAME-SPACE 702
NAME-SPACE 704
748a
748c
748d
AMF
AMF
AMF
NAME-SPACE 712
NAME-SPACE 722
SMF
SMF
SMF
NAME-SPACE 714
NAME-SPACE 724
748b
748e
748f
CORE 742
CORE 744

FIG.7A

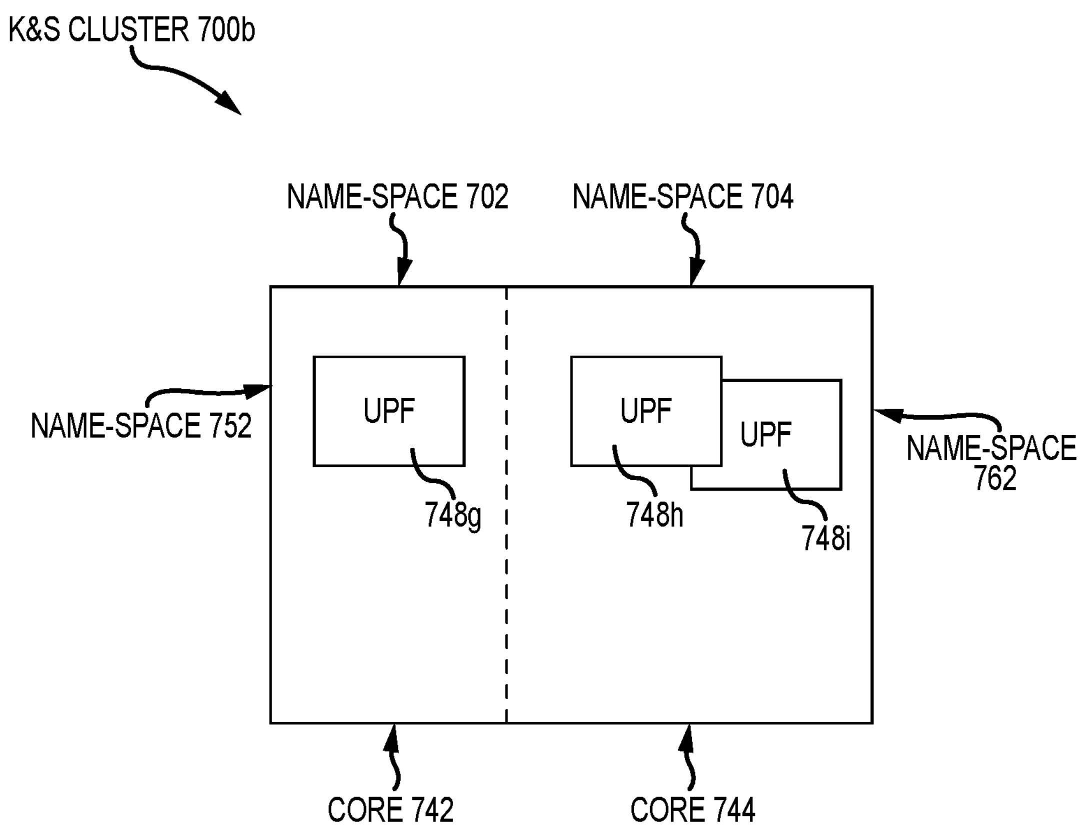
K&S CLUSTER 700b
NAME-SPACE 702
NAME-SPACE 704
UPF
UPF
UPF
NAME-SPACE 752
NAME-SPACE 762
748g
748h
748i
CORE 742
CORE 744

FIG.7B

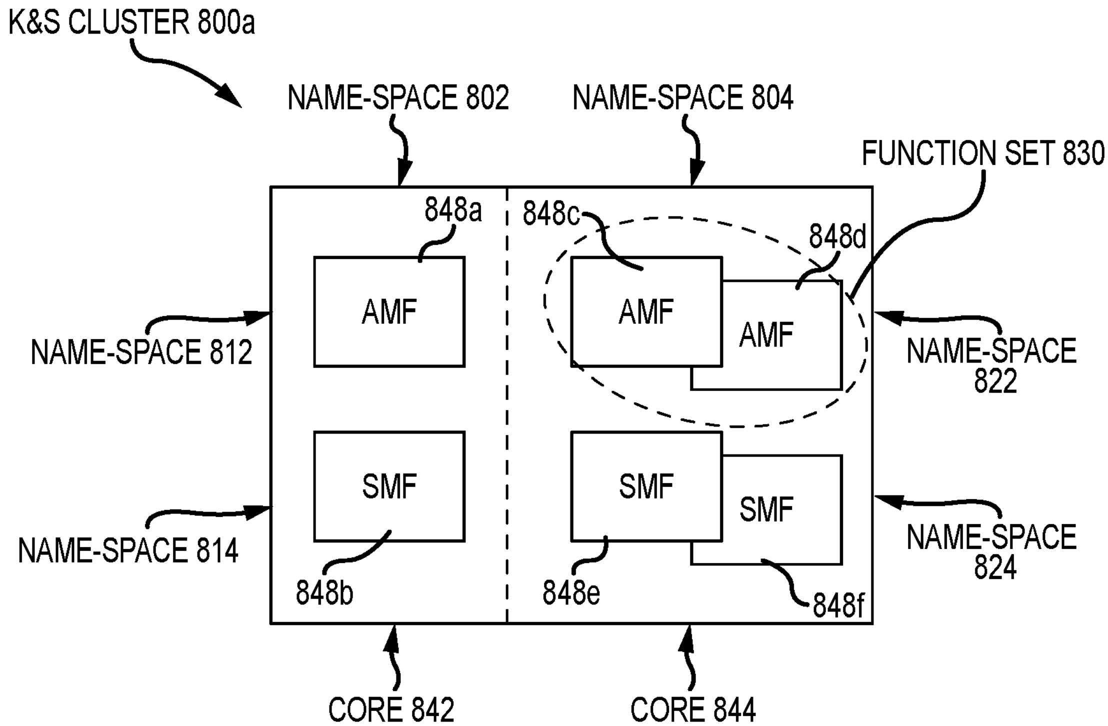
K&S CLUSTER 800a
NAME-SPACE 802
NAME-SPACE 804
FUNCTION SET 830
848a
848c
848d
AMF
AMF
AMF
NAME-SPACE 812
NAME-SPACE 822
SMF
SMF
SMF
NAME-SPACE 814
NAME-SPACE 824
848b
848e
848f
CORE 842
CORE 844

FIG.8A

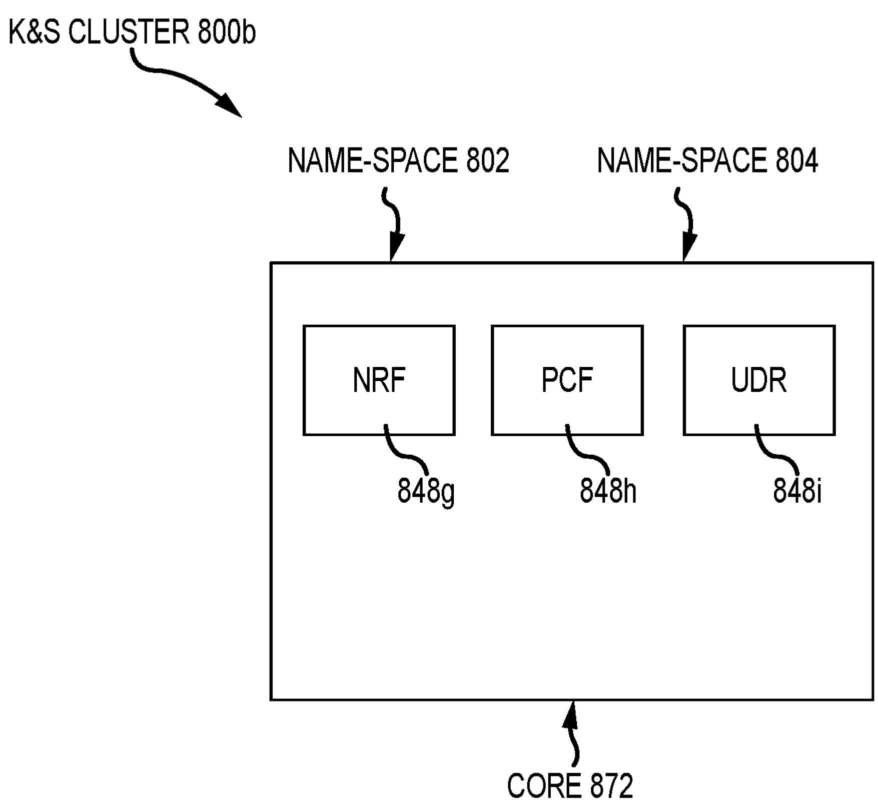
K&S CLUSTER 800b
NAME-SPACE 802
NAME-SPACE 804
NRF
PCF
UDR
848g
848h
848i
CORE 872

FIG.8B

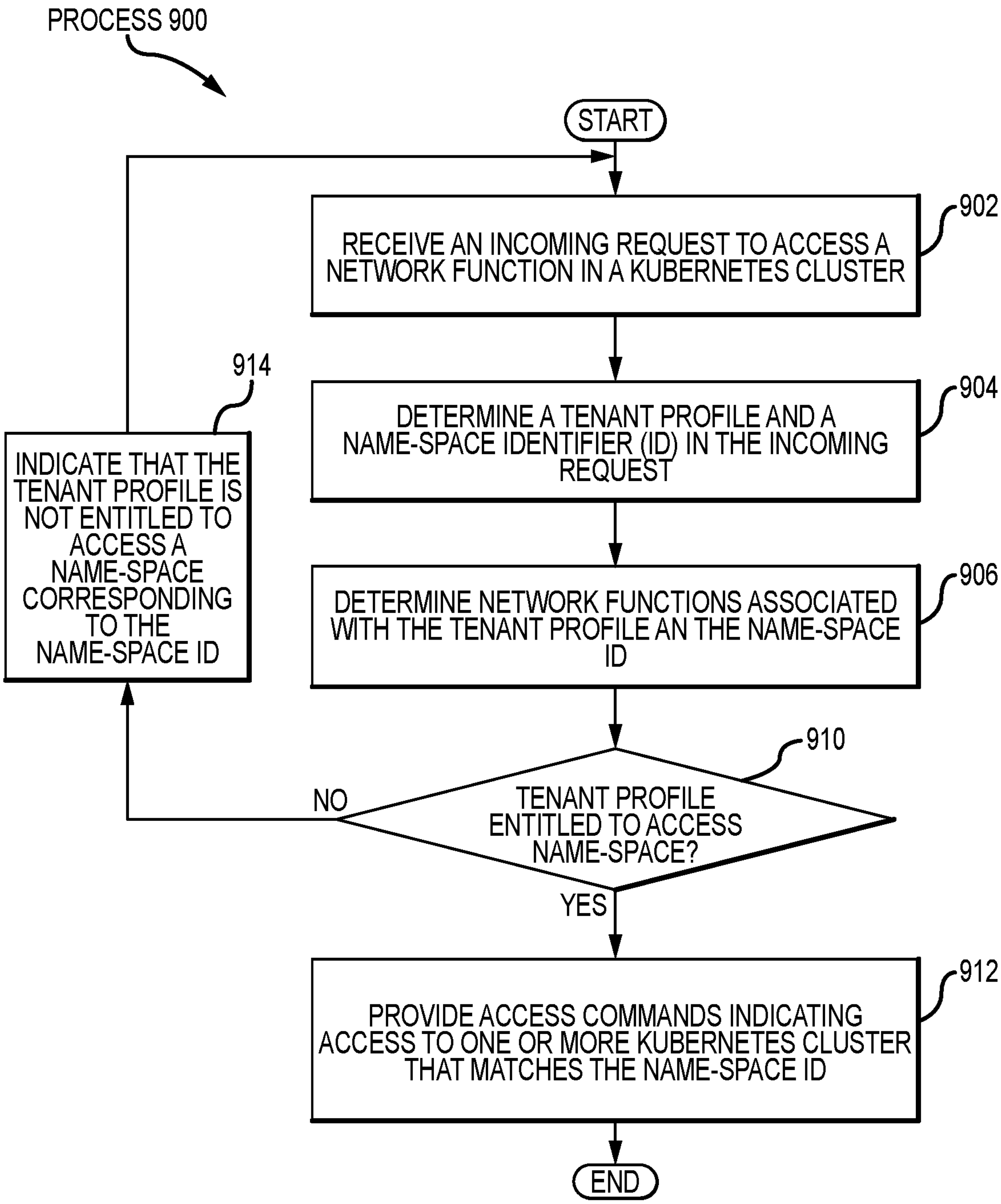
PROCESS 900
START
RECEIVE AN INCOMING REQUEST TO ACCESS A NETWORK FUNCTION IN A KUBERNETES CLUSTER
902
DETERMINE A TENANT PROFILE AND A NAME-SPACE IDENTIFIER (ID) IN THE INCOMING REQUEST
904
DETERMINE NETWORK FUNCTIONS ASSOCIATED WITH THE TENANT PROFILE AN THE NAME-SPACE ID
906
TENANT PROFILE ENTITLED TO ACCESS NAME-SPACE?
910
NO
YES
PROVIDE ACCESS COMMANDS INDICATING ACCESS TO ONE OR MORE KUBERNETES CLUSTER THAT MATCHES THE NAME-SPACE ID
912
END
INDICATE THAT THE TENANT PROFILE IS NOT ENTITLED TO ACCESS A NAME-SPACE CORRESPONDING TO THE NAME-SPACE ID
914

FIG.9

SYSTEM AND METHOD TO MAP HIERARCHICAL MULTI-TENANT ACCESS TO SERVICES

TECHNICAL FIELD

The present disclosure relates generally to access applications in a communication system, and more specifically to a system and method to map hierarchical multi-tenant access to services.

BACKGROUND

In some wireless communications systems, user devices associated with one or more tenants spend several device resources selecting application programming interfaces (API). These device resources may be power resources, memory resources, and processing resources that a given user device consumes while a user attempts to access a new service from the given user device. The device resources are wasted when the given user device lacks a structure to directly access services in a core network. For example, device resources may be wasted by attempting to enter a search query in a browser and scrolling through services to identify the new service to be accessed by the user device. In another example, device resources may be wasted in the process of trying to select multiple services available to the user device.

SUMMARY OF THE DISCLOSURE

Mapping Hierarchical Multi-Tenant Access to Services

In one or more embodiments, the system and method disclosed herein map hierarchical multi-tenant access to services. In particular, the system and method may be configured to map services to specific tenant profiles. Each tenant profile may comprise one or more departments. In accordance with rules and policies associated with a given tenant, the departments associated with the given tenant profile may have access to one or more of the services. Herein, the system and method comprise a hierarchical multi-tenant architecture in which each service may be directly referenced, accessed, or modified in accordance with three different tiers comprising a tenant tier, a department tier, and an application programmable interface (API) tier. In some embodiments, the hierarchical multi-tenant architecture indicates the tenant tier, the department tier, and the API tier in a single application function identifier (ID) (AFID). The AFID may comprise a tenant ID that references a tenant profile associated with a tenant, a department ID that references multiple entitlements associated with a department within the tenant profile, and an API ID that references one or more services associated with the entitlements.

In the hierarchical multi-tenant architecture, a tenant is assigned one tenant identifier (ID) or multiple tenant IDs. The tenant ID may be string of characters comprising symbols, letters, and/or numbers. The tenant ID may comprise human-readable words that indicate a name of a given tenant (e.g., “Tenant1,” “Tenant2,” and the like). A given tenant may comprise multiple departments. Each department may be assigned a department ID (e.g., “Department1,” “Department2,” and the like). Multiple services may be assigned to each department of a tenant. A group of tenants or a group of departments of a tenant may share one or multiple services. For instance, a first tenant ID for a first tenant may be “Tenant1” and a second tenant ID for a second

tenant may be “Tenant2.” For services that may be shared between Tenant1 and Tenant2, a shared tenant ID may be “Tenant1_Tenant2”. In some embodiments, for services that may be shared across Department1 and Department2 of a Tenant1, a shared tenant ID may be “Tenant1.Department1_Department2.” In this regard, access control lists, rules and policies, and system level agreements may be shared across tenants or departments. The tenants and/or corresponding departments may share APIs will be the same for both tenants and/or departments. In this regard, a set of APIs may be mapped to a specific tenant.

In one or more embodiments, the system and method described herein are integrated into a practical application of mapping hierarchical multi-tenant access to services. In this regard, the system and method are configured to map an access to services along with entitlements associated with those services in the AFID because the AFID provides: 1) a specific service; and 2) entitlements enabled by a tenant and one or more departments of the tenant for the specific service.

In addition, the system and method described herein are integrated into a technical advantage of increasing processing speeds in a computer system, because processors associated with the system and method prevent or eliminate waste of resources caused by searching and referencing individual entitlements associated with a request for a given service. Instead, the multi-tenant hierarchical mapping enables the use of the AFID to reference a specific service along with any entitlements available for the specific service in a single information element.

In one or more embodiments, the system and method may be performed by an apparatus, such as a server, communicatively coupled to multiple network components in a core network, one or more base stations in a radio access network, and one or more user equipment. Further, the system may be a wireless communication system, that comprises the apparatus. In addition, the system and method may be performed as part of a process performed by the apparatus communicatively coupled to the network components in the core network. As a non-limiting example, the apparatus may comprise a memory and a processor communicatively coupled to one another. The memory may be configured to store one or more directories comprising access to multiple tenant profiles and one or more network access commands configured to provide access to one or more entitlements. Each tenant profile of the tenant profiles may be associated with one or more services. The processor may be configured to receive a request to access at least one service. The request may comprise an application function identifier (AFID). The tenant ID may reference a tenant profile of the tenant profiles. The department ID may reference multiple entitlements associated with the tenant profile. The API ID may reference a service associated with the entitlements. Further, the processor may be configured to determine multiple network access commands configured to enable access to the service in accordance with the entitlements and generate a report comprising the network access commands.

Implementing Name-Spaces in Hierarchical Multi-Tenant Containerized Service Clusters

In one or more embodiments, the system and method disclosed herein implement name-spaces in hierarchical multi-tenant containerized service clusters. The containerized service clusters may be Kubernetes configured as container orchestration platforms for scheduling and automating deployment, management, and scaling of containerized services (e.g., applications). In particular, the system and method may comprise a multi-core network configured to support services associated with multiple tenants. In this regard, the core network may comprise multiple cores may reside in a multi-cloud environment. The core network may comprise one tenant or multiple tenants. A given tenant may have one or multiple underlying departments. In some embodiments, each core may be mapped to a name-space within one or more Kubernetes (also referred to as K8s) clusters for a given core. As a result, each K8s cluster may comprise have multiple name-spaces. A K8s cluster may comprise multiple nodes in the core network that execute containerized services and applications. A name-space may comprise a containment space or environment created to hold reference, indicator, and/or identifier symbols (i.e. names). An identifier associated with a namespace may be associated only with that namespace.

In some embodiments, a name-space in a K8s cluster may comprise indicators to one or more network functions. A specific network function in the name-space may be accessed by identifying the name-space via a network ID. In cases where the specific network function in a name-space is divided into slice groups, a specific slice group of the specific network function may be accessed by identifying the name-space and the specific slice group in the Network ID. In this regard, a core network may be reached or references via a network ID, network function instance ID, or network slice ID (NSI-ID) that is mapped to a name-space in a specific K8s cluster. Further, the network functions (or sets of network functions of a specific core may be located in different K8s clusters with the same name-space. Herein, the name-space comprises multiple hierarchical accessed which enable different tiers of access. Some tiers may allow access to less network functions in a name-space while other tiers may allow access to more network functions in the same name-space. The network functions associated with a core in a K8s cluster name-space may be replaced or upgraded independently of any network functions located in other name-spaces in other cores. Further, these network functions may be scaled up/down or dimensioned in isolation from network functions in the other cores.

In one or more embodiments, the system and method described herein are integrated into a practical application of implementing name-spaces in hierarchical multi-tenant containerized service clusters. The system and method may be configured to provide access to specific network functions by referencing a name-space location in a core network. The name-space may be referenced and/or accessed using a network ID that is mapped to a hierarchical tier associated with a tenant attempting to access a given network function in the name-space.

In addition, the system and method described herein are integrated into a technical advantage of increasing processing speeds in a computer system, because processors associated with the system and method may directly reference or access network functions that are associated with a given tenant. Further, downtime of the core network may be prevented or eliminated by updating, modifying, or replacing network function in isolation from other network functions in other cores or other name-spaces.

In one or more embodiments, the system and method may be performed by an apparatus, such as a server, communicatively coupled to multiple network components in a core network, one or more base stations in a radio access network, and one or more user equipment. Further, the system may be a wireless communication system, that comprises the apparatus. In addition, the system and method may be performed as part of a process performed by the apparatus communicatively coupled to the network components in the core network. As a non-limiting example, the apparatus may comprise a memory and a processor communicatively coupled to one another. The memory may be configured to store one or more directories comprising access to multiple tenant profiles and one or more network access commands configured to provide access to one or more entitlements. Each tenant profile of the tenant profiles may be associated with one or more network functions. The processor may be configured to receive a request to access at least one network function of the one or more network functions, and extrapolate a tenant profile and a name-space ID from the network ID, or NF instance ID or NSI-ID. The name-space ID may indicate a name-space located in a Kubernetes cluster. Further, the processor may be configured to determine multiple network access commands based at least in part upon the tenant profile and the name-space ID, and generate a report comprising the network access commands. The network access commands may be configured to enable access to the name-space in the Kubernetes cluster.

Certain embodiments of this disclosure may comprise some, all, or none of these advantages. These advantages and other features will be more clearly understood from the following detailed description taken in conjunction with the accompanying drawings and claims.

BRIEF DESCRIPTION OF THE DRAWINGS

For a more complete understanding of this disclosure, reference is now made to the following brief description, taken in connection with the accompanying drawings and detailed description, wherein like reference numerals represent like parts.

FIG. **1** illustrates an example communication system, in accordance with one or more embodiments;

FIGS. **2**A and **2**B illustrate examples of application function identifier (AFID) structures implemented by the communication system of FIG. **1**, in accordance with one or more embodiments;

FIG. **3** illustrates example AFID mapping operations performed by the communication system of FIG. **1**, in accordance with one or more embodiments;

FIG. **4** illustrates an example flowchart of a method to map hierarchical multi-tenant access to services, in accordance with one or more embodiments;

FIGS. **5**A and **5**B illustrate examples of Kubernetes clusters controlled by the communication system of FIG. **1**, in accordance with one or more embodiments;

FIG. **6** illustrates an example of a Kubernetes cluster controlled by the communication system of FIG. **1**, in accordance with one or more embodiments;

FIGS. **7**A and **7**B illustrate examples of Kubernetes clusters controlled by the communication system of FIG. **1**, in accordance with one or more embodiments;

FIGS. **8**A and **8**B illustrate examples of Kubernetes clusters controlled by the communication system of FIG. **1**, in accordance with one or more embodiments; and

FIG. **9** illustrates an example flowchart of a method to implement name-spaces in hierarchical multi-tenant containerized service clusters, in accordance with one or more embodiments.

DETAILED DESCRIPTION

In one or more embodiments, the system and method map hierarchical multi-tenant access to services. In this regard, the system and method map application programming interfaces (API) to one or more departments associated with a given tenant profile in a communication system. One or more services in the communication system may be accessed via an application function identifier (ID) that indicates one or more APIs corresponding to a department within a given tenant. In one or more embodiments, the system and method implement name-spaces in hierarchical multi-tenant containerized service clusters. In this regard, the system and method separate network functions into name-spaces within Kubernetes clusters. A given network function may be accessed by referencing a corresponding name-space.

In one or more embodiments, FIG. **1** illustrates a communication system **100** in which a server **102** generates one or more access commands **104** to access specific services **106** and/or network functions (NFs) **108***a*-**108***h* (collectively, NFs **108**). FIGS. **2**A and **2**B illustrate an application function identifier structure **200***a* and an application function identifier structure **200***b*, respectively. The application function identifier structure **200***a* and the application function identifier structure **200***b* are implemented by the communication system **100** of FIG. **1**. FIG. **3** illustrates application function identifier (ID) (AFID) mapping operations **300** to access the services **106** performed by the communication system **100** of FIG. **1**. FIG. **4** illustrates a process **400** performed by the communication system **100** of FIG. **1**. FIGS. **5**A and **5**B illustrate a K8s cluster **500***a* and a K8s cluster **500***b*, respectively. The K8s cluster **500***a* comprises multiple name-spaces **502**-**524** and the K8s cluster **500***b* comprises one name-space **552** are implemented by the communication system **100** of FIG. **1**. FIG. **6** illustrate a K8s cluster **600** comprising multiple name-spaces **602**-**624** and multiple slices **630**-**640** implemented by the communication system **100** of FIG. **1**. FIGS. **7**A and **7**B illustrate a K8s cluster **700***b* and a K8s cluster **700***b*, respectively. The K8s cluster **700***a* comprises the name-spaces **704**-**724** and the K8s cluster **700***b* comprises the name-spaces **702**, **704**, **752**, and **762** implemented by the communication system **100** of FIG. **1**. FIGS. **8**A and **8**B illustrate a K8s cluster **800***b* and a K8s cluster **800***b*, respectively. The K8s cluster **800***a* comprises the name-spaces **802**-**824** and the K8s cluster **800***b* comprises the name-space **802** are implemented by the communication system **100** of FIG. **1**. FIG. **9** illustrates a process **900** performed by the communication system **100** of FIG. **1**.

Communication System Overview

FIG. **1** illustrates a diagram of a communication system **100** (e.g., a wireless communication system) comprises a server **102** configured to generate network access command **104** to access one or more services **106** and/or one or more network functions (NFs) **108***a*-**108***h* (collectively, NFs **108**), in accordance with one or more embodiments. The services **106** and the NFs **108** may be located in one or more data networks **110** and/or one or more core networks **112**. Herein, the services **106** comprise applications, access to resources, and/or allowance to perform modifications. In FIG. **1**, the server **102** is communicatively coupled to multiple devices in the communication system **100**. While FIG. **1** shows the server **102** connected directly to the one or more data networks **110**, the server **102** may be located inside the core network **112** as part of one or more network components **114***a*-**114***f* (collectively, network components **114**) in the core network **112**.

In one or more embodiments, the communication system **100** comprises the user equipment **116***a*-**116***g* (collectively, user equipment **116**), a radio access network (RAN) **118**, the core network **112**, the one or more data networks **110**, and the server **102**. In some embodiments, the communication system **100** may comprise a Fifth Generation (5G) mobile network or wireless communication system, utilizing high frequency bands (e.g., 24 Gigahertz (GHz), 39 GHz, and the like) or lower frequency bands such (e.g., frequency range FR1 Sub 6 GHz-less than 7.125 GHz). In this regard, the communication system **100** may comprise a large number of antennas. In some embodiments, the communication system may perform one or more communication operations associated with 5G New Radio (NR) protocols described in reference to the Third Generation Partnership Project (3GPP). As part of the 5G NR protocols, the communication system **100** may perform one or more millimeter (mm) wave technology operations to improve bandwidth or latency in wireless communications.

In some embodiments, the communication system **100** may be configured to partially or completely enable communications via one or more various radio access technologies (RATs), wireless communication technologies, or telecommunication standards, such as Global System for Mobiles (GSM) (e.g., Second Generation (2G) mobile networks), Universal Mobile Telecommunications System (UMTS) (e.g., Third Generation (3G) mobile networks), Long Term Evolution (LTE) of mobile networks, LTE-Advanced (LTE-A) mobile networks, 5G NR mobile networks, or Sixth Generation (6G) mobile networks.

Service-Based Architecture

The communication system **100** may comprise a service-based architecture (SBA). The SBA may be an organization scheme in the core network **112** that comprises authentication, security, session management, and aggregation of traffic from end devices (e.g., the user equipment **116**). In the SBA, the core network **112** may be representative of the 5G Core network and comprises multiple network components **114**. In the SBA, the network components **114** are hardware (e.g., electronic circuitry with communication ports, a processor, and a memory) configured to perform one or more specific NFs **108**. Herein, network components **114***a*-**114***f* configured to perform one or more NFs **108** maybe referenced using an NF-associated name. For example, a network component **114***a* configured to perform a network repository function (NRF) **108***a* may be referred to as an NRF (or a NRF network component). In another example, one of the network components **114***a*-**114***f* may comprise a version of the server **102** with a server processor **120** configured to perform one or more specific NFs **108**.

In some embodiments, individual network components **114** provide services or resources to other network components **114** performing different NFs **108**. In other embodiments, each NF is a service provider that allocates one or more resources in communications inside or outside the network components **114** to provide one or more services. The services may be specific for each of the network components **114** and their respective NFs **108** instead of each of the network components **114** providing and consuming processing resources and memory resources to perform multiple NFs **108** in the core network **112**. In 5G NR mobile networks, the SBA is defined by 3GPP to comprise one or more network components **114** configured to perform specific NFs **108** to provide control plane operations and user plane operations. In the 5G NR, the control plane comprises any part of the communication system **100** that controls operations and routing associated with data packets and forwarding operations. Further, in the 5G NR, the user plane comprises any part of the communication system **100** that carries user traffic operations.

In one or more embodiments, the SBA may be configured to provide slices in accordance with specific application scenarios. A slice may be portions of a collection of NFs **108** that are combined into providing specific application resources. The application resources may be provided to one or more user equipment **116** simultaneously via web-based Application Programming Interfaces (APIs). The APIs may enable flexible and agile deployment of innovative services. An API may be a set of instructions that, when executed by a processor, perform modular or cloud-native functions and procedures allowing creation of applications (e.g., the services **106**) that access features or data of an operating system, application, or other service in the communication system **100**.

Communication System Components

Server

The server **102** is generally any device that is configured to process data, communicate with the data networks **110**, one or more network components **114** in the core network **112**, the RAN **118**, and the user equipment **116**. The server **102** may be configured to monitor, track data, control routing of signal, and control operations of certain electronic components in the communication system **100**, associated databases, associated systems, and the like, via one or more interfaces. The server **102** is generally configured to oversee operations of the server processing engine **122**. The operations of the server processing engine **122** are described further below. In some embodiments, the server **102** comprises the server processor **120**, one or more server Input (I)/Output (O) interfaces **124**, a hardware accelerator **126**, and a server memory **128** communicatively coupled to one another. The server **102** may be configured as shown, or in any other configuration. As described above, the server **102** may be located in one of the network components **114** located in the core network **112** and may be configured to perform one or more NFs **108** associated with communication operations of the core network **112**.

In one or more embodiments, the server processor **120**, the server I/O interfaces **124**, the hardware accelerator **126**, and the server memory **128** may be located at a same location or distributed over multiple remote locations separate from one another.

The server processor **120** may comprise one or more processors operably coupled to and in signal communication with the server I/O interfaces **124**, the hardware accelerator **126**, and the server memory **128**. The server processor **120** is any electronic circuitry, including, but not limited to, state machines, one or more central processing unit (CPU) chips, logic units, cores (e.g., a multi-core processor), field-programmable gate arrays (FPGAs), application-specific integrated circuits (ASICs), or digital signal processors (DSPs). The server processor **120** may be a programmable logic device, a microcontroller, a microprocessor, or any suitable combination of the preceding. The one or more processors in the server processor **120** are configured to process data and may be implemented in hardware or software executed by hardware. For example, the server processor **120** may be an 8-bit, a 16-bit, a 32-bit, a 64-bit, or any other suitable architecture. The server processor **120** may comprise an arithmetic logic unit (ALU) to perform arithmetic and logic operations, processor registers that supply operands to the ALU, and store the results of ALU operations, and a control unit that fetches software instructions such as server instructions **130** from the server memory **128** and executes the server instructions **130** by directing the coordinated operations of the ALU, registers and other components via the server processing engine **122**. The server processor **120** may be configured to execute various instructions. For example, the server processor **120** may be configured to execute the server instructions **130** to perform functions or perform operations disclosed herein, such as some or all of those described with respect to FIGS. **1-9**. In some embodiments, the functions described herein are implemented using logic units, FPGAs, ASICs, DSPs, or any other suitable hardware or electronic circuitry.

In the example of FIG. **1**, the server I/O interfaces **124** may comprise one or more displays configured to display a two-dimensional (2D) or three-dimensional (3D) representation of a service. Examples of the representations may comprise, but are not limited to, a graphical or simulated representation of an application, diagram, tables, or any other suitable type of data information or representation. In some embodiments, the one or more displays may be configured to present visual information to one or more users **129**. The one or more displays may be configured to present visual information to the one or more users **129** updated in real-time. The one or more displays may be a wearable optical display (e.g., glasses or a head-mounted display (HMD)) configured to reflect projected images and enable user to see through the one or more displays. For example, the one or more displays may comprise display units, one or more lenses, one or more semi-transparent mirrors embedded in an eye glass structure, a visor structure, or a helmet structure. Examples of display units comprise, but are not limited to, a cathode ray tube (CRT) display, a liquid crystal display (LCD), a liquid crystal on silicon (LCOS) display, a light emitting diode (LED) display, an organic LED (OLED) display, an active-matrix OLED (AMOLED) display, a projector display, or any other suitable type of display. In another embodiment, the one or more displays are a graphical display on the server **102**. For example, the graphical display may be a tablet display or a smartphone display configured to display the data representations.

In one or more embodiments, the server I/O interfaces **124** may be hardware configured to perform one or more communication operations. The server I/O interfaces **124** may comprise one or more antennas as part of a transceiver, a receiver, or a transmitter for communicating using one or more wireless communication protocols or technologies. In some embodiments, the server I/O interfaces **124** may be configured to communicate using, for example, NR or LTE using at least some shared radio components. In other embodiments, the server I/O interfaces **124** may be configured to communicate using single or shared radio frequency (RF) bands. The RF bands may be coupled to a single antenna, or may be coupled to multiple antennas (e.g., for a multiple-input multiple output (MIMO) configuration) to perform wireless communications.

The server I/O interfaces **124** may comprise one or more server network interfaces that may be any suitable hardware or software (e.g., executed by hardware) to facilitate any suitable type of communication in wireless or wired connections. These connections may comprise, but not be limited to, all or a portion of network connections coupled to additional network components **114** in the core network **112**, the RAN **118**, the user equipment **116**, the Internet, an Intranet, a private network, a public network, a peer-to-peer network, the public switched telephone network, a cellular network, a local area network (LAN), a metropolitan area network (MAN), a wide area network (WAN), and a satellite network. The server network interface **124** may be configured to support any suitable type of communication protocol.

The server I/O interfaces **124** may comprise one or more administrator interfaces that may be user interfaces configured to provide access and control to of the server **102** to one or more users **129** via the user equipment **116** or electronic devices. The one or more users **129** may access the server memory **128** upon confirming one or more access credentials to demonstrate that access or control to the server **102** may be modified. In some embodiments, the one or more administrator interfaces may be configured to provide hardware and software resources to the one or more users **129**. Examples of user devices comprise, but are not limited to, a laptop, a computer, a smartphone, a tablet, a smart device, an Internet-of-Things (IoT) device, a simulated reality device, an augmented reality device, or any other suitable type of device. The administrator interfaces may enable access to one or more graphical user interfaces (GUIs) via an image generator display (e.g., the one or more displays), a touchscreen, a touchpad, multiple keys, multiple buttons, a mouse, or any other suitable type of hardware that allow users **129** to view data or to provide inputs into the server **102**. The server **102** may be configured to allow users **129** to send requests to one or more network components **114** or network.

In some embodiments, the hardware accelerator **126** may be any combination of a baseband processor, analog RF signal processing circuitry (e.g., including filters, mixers, oscillators, amplifiers, and the like), or digital processing circuitry (e.g., for digital modulation as well as other digital processing). For example, the hardware accelerator **126** may be configured to allocate power, frequency, and sensing resources during wireless communication operations.

The server memory **128** may be volatile or non-volatile and may comprise a read-only memory (ROM), random-access memory (RAM), ternary content-addressable memory (TCAM), dynamic random-access memory (DRAM), and static random-access memory (SRAM). The server memory **128** may be implemented using one or more disks, tape drives, solid-state drives, and/or the like. The server memory **128** is operable to store the server instructions **130**, one or more requests **132**, one or more directories **134** comprising access to a plurality of tenant profiles **136** associated with the one or more services **106** and the one or more of the NFs **108**, an access control list **138**, one or more rules and policies **140**, one or more access commands **142**, one or more application function IDs (AFID) **144** comprising one or more tenant IDs **146**, one or more department IDs **148**, and one or more application programming interface (API) IDs **150**, one or more system level agreements **152**, one or more Kubernetes (also referred to as K8s) cluster commands **154** configured to control operations associated with one or more K8s clusters **155*a*** and **155*b*** (collectively, K8s clusters **155**), one or more entitlements **156**, one or more reports **158**, and one or more network IDs **160** comprising one or more name-space IDs **162** and one or more slice group IDs **164**. The network IDs **160** may relate the name-space IDs **162** and the slice group IDs **164** to one or more name-spaces or one or more slices. In the server memory **128**, the server instructions **130** may comprise commands and controls for operating one or more specific NFs **108** in the core network **112** when executed by the server processing engine **122** of the server processor **120**.

Herein, the multiple references to K8s clusters are non-limiting examples of containerized service clusters m configured as container orchestration platforms for scheduling and automating deployment, management, and scaling of containerized services (e.g., applications).

In one or more embodiments, the access commands **142** are configured to establish one or more communication sessions between two or more network components **114** in the core network **112**. The access commands **142** may be configured to establish one or more communication sessions between one or more network components **114** in the core network **112** and one of the user equipment **116**. Each configuration command of the access commands **142** may establish a communication session between a first network component of the network components **114** comprising the server **102** and a second network component of the network components **114** based at least in part upon a first configuration command of the access commands **142**. The access commands **142** may be routing and configuration information for reinstating or reestablishing communication sessions when a change is detected in the operations of the core network **112**. For example, in response to losing a specific communication session established with the first access command, the server **102** may attempt to reinstate the specific communication session based at least in part upon a second access command. The access commands **142** may be dynamically or periodically updated from another of the network components **114** in the core network **112**. Herein, communication sessions refer to communication signals exchanged between the server **102** and additional network components **114** in the core network **112**. In some embodiments, the access commands **142** are provided to the server **102** from another of the network components **114** performing a specific NF. The access commands **142** may be configured to enable access of the one or more services **106**. The access commands **142** may be configured to enable access of one or more name-spaces (referenced in FIGS. **4A-8**) and/or one or more slice groups referenced in FIG. **6**) in a given K8s cluster.

The directories **134** may be configured to store service-specific information, tenant-specific information, and/or user-specific information. The directories **134** may enable the server **102** to confirm tenant credentials to access one or more network components (e.g., one of the network components **114** configured to perform the NRF **108*a***, an authentication server function (AUSF) **108*b***, an access and management function (AMF) **108*c***, one or more cloud network functions (CNFs) **108*d***, a policy control function (PCF) **108*e***, a unified data repository (UDR) **108*f***, a session management function (SMF) **108*g***, one or more Service Communication Proxys (SCPs) **108*h***, or the like) in the core network **112**. The directories **134** may be configured to store the tenant profiles **136** and a reference to the one or more services **106**. The directories **134** may be configured to store provider-specific information and service-specific information. The provider-specific information may enable the server **102** to validate credentials associated with a specific provider (e.g., one of the NFs **108**) against corresponding user-specific information and service-specific information.

The requests **132** may be a communication or a message configured to indicate a request for access of an application (via an API) or a service **106**. Further. The entitlements **156** may be configured to provide one or more connectivity allowances (e.g., access) between the server **102**, the user equipment **116**, the base stations **168**, and one or more of the network components **114**. The entitlements **156** may be assigned to specific departments or tenants. The entitlements **156** may be predefined or dynamically defined in accordance with the rules and policies **140**. In the example of FIG. **1**, while the entitlement **156*a*** and the entitlements **156*b*** are shown as part of the entitlements **156**, the entitlements **156** may comprise less or more additional entitlements **156**. The one or more reports **158** may be a communication or a message configured to indicate information to one or more of the network components **114**, the base stations **168**, and/or the user equipment **116**.

The AFIDs **144** may be used for API authentication, service authorization, policies, and one or more system level agreements **152**. The AFID **144** may enable the server **102** to authenticate a given API to specific tenants and one or groups or departments associated with the tenants. The service authorization, the policies, and the system level agreements **152** may be mapped to the tenant IDs **146**, the department ID **148**, and the API ID **150**. The AFID **144** may enable onboarding processes that make mapping of APIs to tenants, and/or departments on the northbound side of a Common API Framework (CAPIF) and Network Exposure Function (NEF) in the core network **112**. On the Southbound side of the NEF, the AFID **144** maps a set of slices to a tenant and/or a department through a slice differentiator (SD) field or information element of a Single Network Slice Selection Assistance Information (S-NSSAI). The SD field may comprise the slice-group ID **164** that indicates a specific tenant ID **146**, department ID **148**, and may comprise priority.

In some embodiments, the AFID **144** is an information element that comprises an availability between 50 characters and 150 characters. The tenant IDs **146** may reference one or more characters indicating a tenant associated with one of the tenant profiles **136**. The department IDs **148** may be configured to reference one or more groups, sub-groups, or portions of a tenant or an organization associated with the tenant. The API IDs **150** may be configured to reference a specific API associated with any given departments of a given tenant. The access control list **138** (also referred to as ACL) may comprise rules that may allow or deny access to one or more of the entitlements **156** (e.g., a virtual environment). The rules and policies **140** may be security configuration commands or regulatory operations predefined by an organization or one or more users **129**. In one or more embodiments, the rules and policies **140** may be dynamically defined by the one or more users **129**. The one or more rules and policies **140** may be one or more a policy as defined in the 3GPP standards. The system level agreements **152** may be configured to define one or more levels of service **106** expected by a tenant, laying out the metrics by which that service **106** is measured, and the remedies or penalties, if any, should the agreed-on service levels not be achieved. The K8s cluster commands **154** may be configuration information and/or commands to control or modify K8s clusters **155** in the cores of the core network **112**.

User Equipment

In one or more embodiments, each of the user equipment **116** may be any computing device configured to communicate with other devices, such as the server **102**, other network components **114** in the core network **112**, databases, and the like in the communication system **100**. Each of the user equipment **116** may be configured to perform specific functions described herein and interact with one or more network components **114** in the core network **112** via one or more base stations **168***a*-**168***g* (collectively, base stations **168**). Examples of user equipment **116** comprise, but are not limited to, a laptop, a computer, a smartphone, a tablet, a smart device, an IoT device, a simulated reality device, an augmented reality device, or any other suitable type of device.

In one or more embodiments, referring to the user equipment **116**A as a non-limiting example of the user equipment **116**, the user equipment **116**A may comprise a user equipment (UE) network interface **170**, a UE I/O interface **172**, a UE processor **174** executing operations via a UE processing engine **176**, and a UE memory **178** comprising one or more instructions **180** configured to be executed by the UE processor **174**. The UE network interface **170** may be any suitable hardware or software (e.g., executed by hardware) to facilitate any suitable type of communication in wireless or wired connections. These connections may comprise, but not be limited to, all or a portion of network connections coupled to additional network components **114** in the core network **112**, the RAN **118**, the Internet, an Intranet, a private network, a public network, a peer-to-peer network, the public switched telephone network, a cellular network, a local area network (LAN), a metropolitan area network (MAN), a wide area network (WAN), and a satellite network. The UE network interface **170** may be configured to support any suitable type of communication protocol.

The UE I/O interface **172** may be hardware configured to perform one or more communication operations. The UE I/O interface **172** may comprise one or more antennas as part of a transceiver, a receiver, or a transmitter for communicating using one or more wireless communication protocols or technologies. In some embodiments, the UE I/O interface **172** may be configured to communicate using, for example, 5G NR or LTE using at least some shared radio components. In other embodiments, the UE I/O interface **172** may be configured to communicate using single or shared RF bands. The RF bands may be coupled to a single antenna, or may be coupled to multiple antennas (e.g., for a MIMO configuration) to perform wireless communications. In some embodiments, the user equipment **116**A may comprise capabilities for voice communication, mobile broadband services (e.g., video streaming, navigation, and the like), or other types of applications. In this regard, the UE I/O interface **172** of the user equipment **116**A may communicate using machine-to-machine (M2M) communication, such as machine-type communication (MTC), or another type of M2M communication.

In some embodiments, the user equipment **116**A is communicatively coupled to one or more of the base stations **168** via one or more communication links **190***a*-**190***g* (e.g., collectively, communication links **190**). The user equipment **116**A may be a device with cellular communication capability such as a mobile phone, a hand-held device, a computer, a laptop, a tablet, a smart watch or other wearable device, or virtually any type of wireless device. In some applications, the user equipment **116** may be referred to as a UE, UE device, or terminal.

The UE processor **174** may comprise one or more processors operably coupled to and in signal communication with the UE network interface **170**, the UE I/O interface **172**, and the UE memory **178**. The UE processor **174** is any electronic circuitry, including, but not limited to, state machines, one or more CPU chips, logic units, cores (e.g., a multi-core processor), FPGAs, ASICs, or DSPs. The UE processor **174** may be a programmable logic device, a microcontroller, a microprocessor, or any suitable combination of the preceding. The one or more processors in the UE processor **174** are configured to process data and may be implemented in hardware or software executed by hardware. For example, the UE processor **174** may be an 8-bit, a 16-bit, a 32-bit, a 64-bit, or any other suitable architecture. The UE processor **174** comprises an ALU to perform arithmetic and logic operations, processor registers that supply operands to the ALU, and store the results of ALU operations, and a control unit that fetches software instructions such as UE instructions **180** from the UE memory **178** and executes the UE instructions **180** by directing the coordinated operations of the ALU, registers, and other components via a UE processing engine **176**. The UE processor **174** may be configured to execute various instructions. For example, the UE processor **174** may be configured to execute the UE instructions **180** to implement functions or perform operations disclosed herein, such as some or all of those described with respect to FIGS. **1-9**. In some embodiments, the functions described herein are implemented using logic units, FPGAs, ASICs, DSPs, or any other suitable hardware or electronic circuitry.

Radio Access Network

In one or more embodiments, the RAN **118** enables the user equipment **116** to access one or more services in the core network **112**. The one or more services may be a mobile telephone service, a Short Message Service (SMS) message service, a Multimedia Message Service (MMS) message service, an Internet access, cloud computing, or other types of data services. The RAN **118** may comprise the base stations **168** in signal communication with the user equipment **116** via the one or more communication links **190**. Each of the base stations **168** may service the user equipment **116***a*-**116***g*. In some embodiments, while multiple base stations **168** are shown connected to multiple user equipment **116** via the communication links **190**, one or more additional base stations **168** may be connected to one or more additional user equipment **116** via one or more additional communication links **190**. For example, the base stations **168***a*-**168***g* may exchange connectivity signals with the user equipment **116***a* via the communication link **190***a*. In another example, the base station **168***g* may exchange connectivity signals with the user equipment **116***g* via the communication link **190***g*. In yet another example, the base stations **168** may service some user equipment **116** located within a geographic area serviced by one of the base

In one or more embodiments, referring to the base station **168***a* as a non-limiting example of the base station **168**, the base station **168***a* may comprise a base station (BS) network interface **182**, a BS I/O interface **184**, a BS processor **186**, and a BS memory **188**. The BS network interface **182** may be any suitable hardware or software (e.g., executed by hardware) to facilitate any suitable type of communication in wireless or wired connections between the core network **112** and the user equipment **116**. These connections may comprise, but not be limited to, all or a portion of network connections coupled to additional network components **114** in the core network **112**, other base stations **168**, the user equipment **116**, the Internet, an Intranet, a private network, a public network, a peer-to-peer network, the public switched telephone network, a cellular network, a LAN, a MAN, a WAN, and a satellite network. The BS network interface **182** may be configured to support any suitable type of communication protocol.

The BS I/O interface **184** may be hardware configured to perform one or more communication operations. The BS I/O interface **184** may comprise one or more antennas as part of a transceiver, a receiver, or a transmitter for communicating using one or more wireless communication protocols or technologies. In some embodiments, the BS I/O interface **184** may be configured to communicate using, for example, 5G NR or LTE using at least some shared radio components. In other embodiments, the BS I/O interface **184** may be configured to communicate using single or shared RF bands. The RF bands may be coupled to a single antenna, or may be coupled to multiple antennas (e.g., for a MIMO configuration) to perform wireless communications. In some embodiments, the base station **168***a* may allocate resources in accordance with one or more routing and configuration operations obtained from the core network **112**. In some embodiments, resources may be allocated to enable capabilities in the user equipment **116** for voice communication, mobile broadband services (e.g., video streaming, navigation, and the like), or other types of applications.

In some embodiments, the base station **168***a* is communicatively coupled to one or more of the user equipment **116** via the one or more communication links **190**. In some applications, the base stations **168** may be referred to as a BS, evolved Node B (eNodeB or eNB), a next generation Node B, gNodeB, gNB, or terminal.

The BS processor **186** may comprise one or more processors operably coupled to and in signal communication with the BS network interface **182**, the BS I/O interface **184**, and the BS memory **188**. The BS processor **186** is any electronic circuitry, including, but not limited to, state machines, one or more CPU chips, logic units, cores (e.g., a multi-core processor), FPGAs, ASICs, or DSPs. The BS processor **186** may be a programmable logic device, a microcontroller, a microprocessor, or any suitable combination of the preceding. The one or more processors in the BS processor **186** are configured to process data and may be implemented in hardware or software executed by hardware. For example, the BS processor **186** may be an 8-bit, a 16-bit, a 32-bit, a 64-bit, or any other suitable architecture. The BS processor **186** comprises an ALU to perform arithmetic and logic operations, processor registers that supply operands to the ALU, and store the results of ALU operations, and a control unit that fetches software instructions (not shown) from the BS memory **188** and executes the software instructions by directing the coordinated operations of the ALU, registers, and other components via a processing engine (not shown) in the BS processor **186**. The BS processor **186** may be configured to execute various instructions. For example, the BS processor **186** may be configured to execute the software instructions to implement functions or perform operations disclosed herein, such as some or all of those described with respect to FIGS. **1-9**. In some embodiments, the functions described herein are implemented using logic units, FPGAs, ASICs, DSPs, or any other suitable hardware or electronic circuitry.

Core Network

The core network **112** may be a network configured to manage communication sessions for the user equipment **116**. In one or more embodiments, the core network **112** may establish connections between user equipment **116** and a particular data network **110** in accordance with one or more communication protocols. As it will be described in reference to FIGS. **5A-9**, the core network **112** may be a multi-core network **112** configured to comprise multiple cores. In this regard, the multi-core network may comprise multiple NFs **108** in each core. In the example of FIG. **1**, the core network **112** comprises the network component **114***a* configured to perform the NRF **108***a*, the network component **114***b* configured to perform the AUSF **108***b*, the network component **114***c* configured to perform the AMF **108***c*, the network component **114***d* configured to perform the CNFs **108***d*, the network component **114***e* configured to perform the PCF **108***e* and the UDR **108***f*, and the network component **114***f* configured to perform the SMF **108***g* and the SCPs **108***h*. Herein, as a non-limiting example, while the NRF **108***a* is associated with the network component **114***a*, the core network **112** may comprise multiple network component **114** performing the NRF **108***a*. For example, a Unified Data Management (UDM) may be part of a core.

In some embodiments, the NRF **108***a* may comprise a service registration procedure that accesses the one or more databases to store or retrieve routing and configuration information associated with one or more network components **114** in the core network **112**. The NRF **108***a* may access the database to discover services offered by other networks or other network components **114** with service discovery procedures and service authorization procedures. The NRF **108***a* may maintain a list of available NFs operations available in the core network **112** and any network components **114** associated with performing a given NF **108**. The NRF **108***a* may also performs registration and discovery of service such that different NFs **108** may find each other via APIs. As an example, when the SMF **108***g* is registered to the NRF **108***a*, the SMF **108***g* is discoverable by the AMF **108***c* when the user equipment **116** attempts to access a given service type via the SMF **108***g*. In other embodiments, the NFs **108** may be connected via a communication bus to all other additional network elements in the core network **112**. In the SBA, the NRF **108***a* may enable access between the user equipment **116** and the services offered via the NFs **108**.

In one or more embodiments, the network components **114***d* performing the one or more CNFs **108***d* may be configured to operate multiple services associated with one or more services **106**, while dynamically directing network traffic within the core network **112**. In some embodiments, the network component **114***f* performing the SMF **108***g* may be configured to manage one or more communication sessions established between network components **114** of the core network **112**, allocate and manage resource allocation routing for the user equipment **116**, user plane selection, QoS and configuration enforcements for the control plane, service registration, discovery, establishment, and the like. In other embodiments, the network component **114***c* performing the AMF **108***c* may be configured to manage mobility, registration, connections, and overall access for the other network components **114** in the core network **112**. The AMF **108***c* may act as an entry point for connections between the user equipment **116** and a given service. In yet other embodiments, the network component **114***f* performing the one or more SCPs **108***h* may be configured to provide a point of entry for a cluster of NFs **108** in the core network **112** to the user equipment **116** once the user equipment **116** are discovered by the NRF **108***a*. This allows the SCPs **108***h* to be delegated discovery points in the core network **112**. The network component **114***b* performing the AUSF **108***b* may be configured to share performing of some of the aforementioned operations with a Unified Data Management (UDM) (not shown). In this regard, the AUSF **108***b* may be configured to perform authentication processes while the UDM manages user data for any other processes in the core network **112**. In other embodiments, the UDM may receive requests for subscriber data from the SMF **108***g*, the AMF **108***c*, and the AUSF **108***b* before providing any services **106**. The AUSF **108***b* may be implemented in one of the network components **114** configured to enable the AMF **108***c* to authenticate the user equipment **116**. The network component **114***e* performing the PCF **108***e* may be configured to provide a policy control framework in which the rules and policies **140** are implemented in accordance with one or more application guidelines. In some embodiments, the PCF **108***e* may apply policy decisions to services provided, accessing subscription information, and the like to control behavior associated with the core network **112**. The network component **114***f* performing the UDR **108***f* configured to operate as a centralized data repository for subscription data, subscriber policy data, session information, context information, and application states. In some embodiments, the UDR **108***f* may be configured to provide API integrations with other NFs **108** to retrieve subscriber subscription and policy data. The UDR **108***f* may notify other NFs **108** of changes in subscriber data, supports real-time or batch (e.g., bulk) data access provisioning and subscriber data provisioning, and manages service parameters and application data for advanced applications.

In some embodiments, the core network **112** enables the user equipment **116** to communicate with the server **102**, or another type of device, located in a particular data network **110** or in signal communication with a particular data network **110**. The core network **112** may implement a communication method that does not require the establishment of a specific communication protocol connection between the user equipment **116** and one or more of the data networks **110**. The core network **112** may include one or more types of network devices (not shown), which may perform different NFs **108**.

In some embodiments, the core network **112** may include a 5G NR or an LTE access network (e.g., an evolved packet core (EPC) network) among others. In this regards, the core network **112** may comprise one or more logical networks implemented via wireless connections or wired connections. Each logical network may comprise an end-to-end virtual network with dedicated power, storage, or computation resources. Each logical network may be configured to perform a specific application comprising individual policies, rules, or priorities. Further, each logical network may be associated with a particular Quality of Service (QoS) class, type of service, or particular user associated with one or more of the user equipment **116**. For example, a logical network may be a Mobile Private Network (MPN) configured for a particular organization. In this example, when the user equipment **116***a* is configured and activated by a wireless network associated with the RAN **118**, the user equipment **116***a* may be configured to connect to one or more particular network slices (i.e., logical networks) in the core network **112**. Any logical networks or slices that may be configured for the user equipment **116***a* may be configured using one of the network components **114** of FIG. **1** performing a Network Slice Selection Function (NSSF) that may store a subscription profile associated with the user equipment **116***a*, in a network component operating as a Unified Data Management (UDM). Further, when the user equipment **116***a* may request a connection to a particular logical network or slice, the user equipment **116***a* may send a request to the network component performing the AMF **108***c*. The AMF **108***c* may provide a list of allowed logical networks or slices to the user equipment **116***a*. The user equipment **116***a* may then request a Packet Data Unit (PDU) connection with one or more of the provided logical networks or slices.

Data Networks

In the example system **100** of FIG. **1**, the data networks **110** may facilitate communication within the communication system **100**. This disclosure contemplates that the data networks **110** may be any suitable network operable to facilitate communication between the server **102**, the core network **112**, the RAN **118**, and the user equipment **116**. The data networks **110** may include any interconnecting system capable of transmitting audio, video, signals, data, messages, or any combination of the preceding. The data networks **110** may include all or a portion of a LAN, a WAN, an overlay network, a software-defined network (SDN), a virtual private network (VPN), a packet data network (e.g., the Internet), a mobile telephone network (e.g., cellular networks, such as 4G or 5G), a Plain Old Telephone (POT) network, a wireless data network (e.g., WiFi, WiGig, WiMax, and the like), a Long Term Evolution (LTE) network, a Universal Mobile Telecommunications System (UMTS) network, a peer-to-peer (P2P) network, a Bluetooth network, a Near Field Communication network, a Zigbee network, or any other suitable network, operable to facilitate communication between the components of the communication system **100**. In other embodiments, the communication system **100** may not have all of these components or may comprise other elements instead of, or in addition to, those above.

Application Function Identifier Structures

FIGS. **2**A and **2**B illustrate example structures associated with the AFID in accordance with one or more embodiments. In the example of FIG. **2**A, an AFID structure **200***a* is shown comprising a number of AFID characters divided into a number of M characters corresponding to a tenant ID **202**, a number of N characters corresponding to a department ID **204**, and a number of K characters corresponding to an API ID **206**. In the example of FIG. **2**B, an AFID structure **200***b* is shown comprising a number of AFID characters divided into a number of P characters corresponding to a tenant ID **252**, a number of Q characters corresponding to a department ID **254**, and a number of O characters corresponding to an API ID **256**. Further, the length of each field may be variable by using any character or symbol (e.g., dash symbol, dot character, space, underscore character or the like) between the fields (e.g., TENANT1-DEPARTMENT6-API7"). In some embodiments, the length of each field may be modified by using abbreviation, short phrases, or nicknames for the tenant IDs **146**, the department IDs **148**, and/or the API IDs **150** (e.g., "TEN1.DEPT1.API1" using nickname "TENT1" for a specific tenant ID **146** and abbreviation "DEPT1" for a specific department ID **148**).

In FIG. **2**A, as a non-limiting representative example, the number of AFID characters in the AFID structure **200***a* may be 100 characters, the number of M characters may be 50 characters, the number of N characters may be 35 characters, and the number of K characters may be 15 characters. The number of M characters, the number of N characters, and the number of K characters may be modified such that they comprise different values that, when added together, remain within a total of 100 characters available in the AFID characters.

In FIG. **2**B, as a non-limiting representative example, the number of AFID characters in the AFID structure **200***b* may be 120 characters, the number of P characters may be 50 characters, the number of Q characters may be 40 characters, and the number of O characters may be 30 characters. The number of P characters, the number of Q characters, and the number of O characters may be modified such that they comprise different values that, when added together, remain within a total of 120 characters available in the AFID characters.

In some embodiments, the number of AFID characters may be modified in accordance with the rules and policies **140**. For example, the number of AFID characters may be variable to be between 200 characters and 3 characters, inclusive. The number of AFID characters may be a number of available characters. In this regard, the AFID may comprise less characters than those available when referencing a specific API. For example, the AFID "TENANT6.DEPARTMENT2.API9" may be mapped to reference or access an API named "API9" to be operated in accordance with entitlements granted to a department named "DEPARTMENT2" of a tenant named "TENANT6." In this example, while the number of characters in the AFID "TENANT6.DEPARTMENT2.API9" is equal to 22 characters (excluding the periods), the number of AFID characters available may remain 100 characters. Further, the number of characters occupied by tenant IDs **146**, the department IDs **148**, and the API IDs **150** in a given AFID may be different or equal to one another. For example, a number of AFID characters may be 90 characters with 30 characters corresponding to each of the tenant IDs **146**, the department IDs **148**, and the API IDs **150**. In another example, a number of AFID characters may be 170 characters with 50 characters corresponding to the tenant IDs **146**, 100 characters corresponding to the department IDs **148**, and 20 characters corresponding to the API IDs **150**.

Application Function Identifier Mappings Operations

FIG. **3** illustrates examples of multiple AFID mapping operations **300**, in accordance with one or more embodiments. The AFID mapping operations **300** are shown in a table form where example entitlements **156***a*-**156***d* to multiple AFIDs are mapped to AFIDs **310**-**320**. Each AFID **144** of the AFIDs **310**-**320** is divided into a Tenant-ID, a Department-ID, and an API-ID corresponding to one or more entitlements example **156***a*-**156***d*.

As a non-limiting example, the entitlement **156***a* comprises access to monitoring information, the entitlement **156***b* comprises event controls, the entitlement **156***c* comprises directory management, and the entitlement **156***d* comprises system access. As described above, the entitlements **156***a*-**156***d* may be accessed by one or more tenants, one or more departments, and one or more APIs. The entitlements **156***a*-**156***d* may be unique to each API. For example, one API may comprise a first version of the entitlement **156***a* and another API may comprise a second version of the entitlement **156***a*. Further, the entitlements **156***a*-**156***d* may be different depending on the tenant ID **146** and/or the department ID **148**. In the example of FIG. **3**, the server **102** may receive a first request with an AFID **310** comprising "TENANT1.DEPARTMENT1.APPLICATION1" requesting access to a first API in accordance with the entitlements **156** for a first department of a first tenant. The entitlements **156** may comprise access to the entitlement **156***a* for the first department of the first tenant. The server **102** may receive a second request with an AFID **312** comprising "TENANT1.DEPARTMENT2.APPLICATION2" requesting access to a second API in accordance with the entitlements **156** for a second department of the first tenant. The entitlements **156** may comprise access to the entitlement **156***b* and the entitlement **156***c* for the second department of the first tenant. The server **102** may receive a request with an AFID **314** comprising "TENANT1.DEPARTMENT2.APPLICATION3" requesting access to a third API in accordance with the entitlements **156** for the second department of the first tenant. The entitlements **156** may comprise access to the entitlement **156***a* and the entitlement **156***d* for the second department of the first tenant. The server **102** may receive a request with an AFID **316** comprising "TENANT1.DEPARTMENT3.APPLICATION3" requesting access to the third API in accordance with the entitlements **156** for the third department of the first tenant. The entitlements **156** may comprise access to the entitlement **156***a* and the entitlement **156***d* for the third department of the first tenant. Further, the server **102** may receive a request with an AFID **320** comprising "TENANT2.DEPARTMENT1. APPLICATION1" requesting access to a first API in accordance with the entitlements **156** for a first department of a second tenant. The entitlements **156** may comprise access to the entitlement **156***a* for the first department of the second tenant.

In one or more embodiments, for a tenant with no departments, the department ID **148** may be set to "NULL" (e.g., code "0x00," code "0," a configured character symbol, or an empty space). In some embodiments, all the departments may be selected in a specific tenant. In this regard, the department ID **148** may be set to "ALL," a specific symbol (e.g., code "FxFF", or a configured character symbol "*") to reference all the department IDs **148** associated with a specific tenant. As a non-limiting example in reference to FIG. **3**, all department entitlements for the first tenant may be selected in the AFID **144** set to "TENANT1.ALL.API1" for an API1. In some embodiments, this AFID **144** structure may help to map an API to an appropriate tenant and/or department during an onboarding process. In other embodiments, the architecture may use the tenant IDs **146** and the API IDs **150** only without the department IDs **148**.

In one or more embodiments, the example of FIG. **3** comprises an access control list **133** in which event type(s) associated with an API are mapped hierarchically in accordance with the tiers provided by entitlements at an API level, a department level, and a tenant level. One column may represent the AFID **144** and the other columns represent event types such as those described in technical specifications of the 3GPP standards. After proper authentication, relevant personnel of a tenant may obtain access to the fields associated with the event types in the access control list **133**. Similar to the example of FIG. **3**, AFIDs **144** may be mapped to the rules and policies **140** (e.g., policy controls as defined in the 3GPP standards) and the system level agreements **152**.

Example Process to Map Hierarchical Multi-Tenant Access to Services

FIG. **4** illustrates an example flowchart of a process **400** to map hierarchical multi-tenant access to the services, in accordance with one or more embodiments. In one or more embodiments, the process **400** comprises operations **402**-**414**. Modifications, additions, or omissions may be made to the process **400**. The process **400** may include more, fewer, or other operations than those shown below. For example, operations may be performed in parallel or in any suitable order. While at times discussed as the server **102**, one or more of the user equipment **116**, components of any of thereof, or any suitable system or components of the communication system **100** may perform one or more operations of the process **400**. For example, one or more operations of the process **400** may be implemented, at least in part, in the form of server instructions **130** of FIG. **1**, stored on non-transitory computer readable media, tangible media, machine-readable media (e.g., server memory **128** of FIG. **1** operating as a non-transitory computer readable medium) that when run by one or more processors (e.g., the server processor **120** of FIG. **1**) may cause the one or more processors to perform operations described in operations **402**-**414** of the process **400**.

The process **400** starts at operation **402**, where the server **102** receives an incoming request **132** to access one or more services **106**. At operation **304**, the server **102** determines an AFID **144** in the incoming request **132**. The AFID **144** may comprise multiple characters corresponding to a tenant ID **146**, multiple characters corresponding to an department ID **148**, and multiple characters corresponding to an API ID **150**. As described above, the AFID **144** is an information element that comprises an availability between 50 characters and 150 characters. At operation **306**, the server **102** extrapolates the tenant ID **146**, the department ID **148**, and the API ID **150** from the AFID **144**.

The process **400** continues at operation **410**, where the server **102** is configured to determine whether the tenant ID **146**, the department ID **148**, and the API ID **150** match information in the directories **134**. If the server **102** determines that the identifiers does not match information in the directories **134** (i.e., NO), the process **400** proceeds to operation **412**. At operation **412**, the server **102** indicates that the AFID **144** does not match the information in the directories **134** as an alert. The alert may be a visual alert or a sound alert presented to one or more users **129** via a corresponding user equipment **116**. If the server **102** determines that the identifiers match information in the directories **134** (i.e., YES), the process **400** proceeds to operation **414**.

In this case, the process **400** may conclude at operation **414**, where the server **102** provides network access commands **142** indicating access to one or more entitlements **156** that match the tenant ID **146**, the department ID **148**, and the API ID **150**. The server **102** may be configured to generate a report **158** (e.g., a signal or a communication) indicating or comprising information indicating the network access commands **142**. In this regard, the server **102** may present the report **158** to a user equipment **116** configured to access the service **106** based at least in part upon the network access commands **142** in the report **158**.

Name-Space for Multi-Tenancy

FIGS. **5**A and **5**B show examples of K8s clusters **155**, in accordance with one or more embodiments. FIG. **5**A shows a K8s cluster **500***a* comprising multiple name-spaces **502**-**524** and multiple cores **542** and **544**. FIG. **5**B shows a K8s cluster **500***b* comprising one name-space **552** and a core **572**.

In the example of FIG. **5**A, as a non-limiting representative example, the K8s cluster **500***a* comprises multiple NFs **108** separated into a core **542** and a core **544**. In FIG. **5**A, each of the name-spaces **502**-**524** is mapped to the core **542** or the core **544**. In some embodiments, the name-space IDs **162** may reference one of the name-spaces **502**-**524** to access one NF **108** or multiple NFs **108** (e.g., a function set **530**). The core **542** comprises a name-space **502**, a name-space **512**, and a name-space **514**. The name-space **502** may be referenced to access an AMF **548***a* and a SMF **548***b* in the core **542**, the name-space **512** may be referenced to access the AMF **548***a* in the core **542**, and the name-space **514** may be referenced to access the SMF **548***b* in the core **542**. In some embodiments, the network access commands **142** may be configured to provide access to the AMF **548***a* and/or the SMF **548***b* via the name-space **502**, the name-space **512**, and the name-space **514**. The core **544** comprises a name-space **504**, a name-space **522**, and a name-space **524**. The name-space **504** may be referenced to access an AMF **548***c*, an AMF **548***d*, a SMF **548***e*, and a SMF **548***f* in the core **544**, the name-space **522** may be referenced to access the AMF **548***c* and the AMF **548***d* in the core **544**, and the name-space **524** may be referenced to access the SMF **548***e* and the SMF **548***f* in the core **544**. In some embodiments, the network access commands **142** may be configured to provide access to the AMF **548***c*, the AMF **548***d*, the SMF **548***e*, and/or the SMF **548***f* via the name-space **504**, the name-space **522**, and the name-space **524**. Further, the network access commands **142** may be configured to provide access to the AMF **548***c* and/or the AMF **548***d* in the function set **530**.

In the example of FIG. **5**B, as a non-limiting representative example, the K8s cluster **500***b* comprises multiple NFs **108** in a single core **572**. In FIG. **5**B, the name-space **552** is mapped to the core **572**. In some embodiments, the name-space IDs **162** may reference the name-space **552** to access a portion of the NFs **108** (e.g., a function set **560**) or all the

NFs **108**. The name-space **552** may be referenced to access an AMF **548***g*, an AMF **548***h*, an AMF **548***i*, a SMF **548***j*, a SMF **548***k*, and a SMF **548***l* in the core **572**. In some embodiments, the network access commands **142** may be configured to provide access to the AMF **548***g*, the AMF **548***h*, the AMF **548***i*, the SMF **548***j*, the SMF **548***k*, and the SMF **548***l* in the core **572** via the name-space **552**. Further, the network access commands **142** may be configured to provide access to the AMF **548***g*, the AMF **548***h*, and/or the AMF **548***i* in the function set **560**.

In one or more embodiments, while the K8s cluster **500***a* and the K8s cluster **500***b* show certain NFs **108**, the K8s cluster **500***a* and the K8s cluster **500***b* may comprise less or more NFs **108**. The AMF **548***a*, the AMF **548***c*, the AMF **548***d*, and the AMFs **548***g*-**548***i* may perform one or more operations similar to those described in reference to the AMF **108***c* of FIG. **1**. The SMF **548***b*, the SMF **548***e*, the SMF **548***f*, and the SMFs **548***j*-**5481** may perform one or more operations similar to those described in reference to the SMF **108***g* of FIG. **1**.

Name-Space and Slice Group for Hierarchical Multi-Tenancy

FIG. **6** show an example of a K8s cluster **155**, in accordance with one or more embodiments. FIG. **6** shows a K8s cluster **600** comprising multiple name-spaces **602**-**624**, multiple slice groups in slices **630**-**640**, and multiple cores **652** and **654**. In the example of FIG. **6**, as a non-limiting representative example, the K8s cluster **600** comprises multiple NFs **108** separated into a core **652** and a core **654**. In FIG. **6**, each of the name-spaces **602**-**624** is mapped to the core **652** or the core **654**. In some embodiments, the name-space IDs **162** may reference one of the name-spaces **602**-**624** to access one NF **108** or multiple NFs **108**.

The core **652** comprises a name-space **602**, a name-space **612**, and a name-space **614**. Further, the core **652** comprises a slice **630** and a slice **632**. The name-space **602** may be referenced to access the slice **630** or the slice **632** in an AMF **648***a* and a SMF **648***b* in the core **652**, the name-space **612** may be referenced to access the slice **630** or the slice **632** in the AMF **648***a* in the core **652**, and the name-space **614** may be referenced to access the slice **630** or the slice **632** in the SMF **648***b* in the core **652**. In some embodiments, the network access commands **142** may be configured to provide access to the AMF **648***a* and/or the SMF **648***b* via the name-space **602**, the name-space **612**, and the name-space **614**. One or more specific operations associated with a given NF **108** may be accessed via the slice **630** and the slice **632**. As a non-limiting example, the slice **630** may be configured in accordance with entitlements **156** for a Slice-Group-1 that may be assigned to a first department of a first tenant and the slice **632** may be configured in accordance with entitlements **156** for a Slice-Group-2 that may be assigned to a second department of a second tenant.

The core **654** comprises a name-space **604**, a name-space **622**, and a name-space **624**. Further, the core **654** comprises a slice **634**, a slice **636**, and a slice **640**. The name-space **604** may be referenced to access the slice **634**, the slice **636**, and the slice **640** in an AMF **648***c*, an AMF **648***d*, a SMF **648***e*, and a SMF **648***f* in the core **654**, the name-space **622** may be referenced to access the slice **634**, the slice **636**, or the slice **640** in the AMF **648***c* and the AMF **648***d* in the core **654**, and the name-space **624** may be referenced to access the slice **634**, the slice **636**, or the slice **640** in the SMF **648***e* and the SMF **648***f* in the core **654**. In some embodiments, the network access commands **142** may be configured to provide access to the slice **634**, the slice **636**, and the slice **640** of the AMF **648***c*, the AMF **648***d*, the SMF **648***e*, and/or the SMF **648***f* via the name-space **604**, the name-space **622**, and the name-space **624**. One or more specific operations associated with a given NF **108** may be accessed via the slice **634**, the slice **636**, and the slice **640**. As a non-limiting example, the slice **634** may be configured in accordance with entitlements **156** for a Slice-Group-3 that may be assigned to a first department of the second tenant, the slice **636** may be configured in accordance with entitlements **156** for a Slice-Group-4 that may be assigned to the second department of the second tenant, and the slice **640** may be configured in accordance with entitlements **156** for a Slice-Group-5 that may be assigned to a third department of the second tenant.

Multi-Tenancy in Same Name-Spaces with Multiple Kubernetes Clusters

FIGS. 7A and 7B show examples of K8s clusters **155**, in accordance with one or more embodiments. FIG. **7**A shows a K8s cluster **700***a* comprising multiple name-spaces **702**-**724** and multiple cores **742** and **744**. FIG. **7**B shows a K8s cluster **700***b* comprising multiple name-spaces **702**, **704**, **752**, and **754** and the same cores **742** and **744** of the FIG. **7**A.

In the example of FIG. 7A, as a non-limiting representative example, the K8s cluster **700***a* comprises multiple NFs **108** separated into a core7 and a core **744**. In FIG. 7A, each of the name-spaces **702**-**724** is mapped to the core **742** or the core **744**. In some embodiments, the name-space IDs **162** may reference one of the name-spaces **702**-**724** to access one NF **108** or multiple NFs **108**. The core **742** comprises a name-space **702**, a name-space **712**, and a name-space **714**. The name-space **702** may be referenced to access an AMF **748***a* and a SMF **748***b* in the core **742**, the name-space **712** may be referenced to access the AMF **748***a* in the core **742**, and the name-space **714** may be referenced to access the SMF **748***b* in the core **742**. In some embodiments, the network access commands **142** may be configured to provide access to the AMF **748***a* and/or the SMF **748***b* via the name-space **702**, the name-space **712**, and the name-space **714**. The core **744** comprises a name-space **704**, a name-space **722**, and a name-space **724**. The name-space **704** may be referenced to access an AMF **748***c*, an AMF **748***d*, a SMF **748***e*, and a SMF **748***f* in the core **744**, the name-space **722** may be referenced to access the AMF **748***c* and the AMF **748***d* in the core **744**, and the name-space **724** may be referenced to access the SMF **748***e* and the SMF **748***f* in the core **744**. In some embodiments, the network access commands **142** may be configured to provide access to the AMF **748***c*, the AMF **748***d*, the SMF **748***e*, and/or the SMF **748***f* via the name-space **704**, the name-space **722**, and the name-space **724**.

In the example of FIG. 7B, as a non-limiting representative example, the K8s cluster **700***b* comprises multiple NFs **108** separated into the core **742** and the core **744**. In FIG. 7B, the name-space **752** is mapped to the core **772**. In some embodiments, the name-space IDs **162** may reference the name-space **702**, the name-space **704**, the name-space **752**, and the name-space **762** to access a portion of the NFs or all the NFs **108**. The name-space **702** may be referenced to access an AMF **548***g*, an AMF **548***h*, an AMF **548***i*,AMF **748***a*, the SMF **748***b*, and a UPF **748***g* in the core **742**, the name-space **704** may be referenced to access the AMF **748***c*, the AMF **748***d*, the SMF **748***e*, the SMF **748***f*, a UPF **748***h*, and a UPF **748***i* in the core **742**, the name-space **752** may be referenced to access the UPF **748***g* in the core **742**, and name-space **762** may be referenced to access the UPF **748***h* and the UPF **748***i* in the core **744**.

In one or more embodiments, while the K8s cluster **700***a* and the K8s cluster **700***b* show certain NFs **108**, the K8s cluster **700***a* and the K8s cluster **700***b* may comprise less or more NFs **108**. The AMF **748***a*, the AMF **748***c*, and the AMF **748***d* may perform one or more operations similar to those described in reference to the AMF **108***c* of FIG. **1**. The SMF **748***b*, the SMF **748***e*, and the SMF **748***f* may perform one or more operations similar to those described in reference to the SMF **108***g* of FIG. **1**. The UPF **748***g*, the UPF **748***h*, and the UPF **748***i* may be configured to connect data coming over the RAN **118** to the data networks **110**. In some embodiments, the UPF **222**, UPF **748***g*, the UPF **748***h*, and the UPF **748***i* may be able to route data packets to correct destinations on the data networks **110**.

Shared Network Functions in One or More Cores

FIGS. **8**A and **8**B show examples of K8s clusters **155**, in accordance with one or more embodiments. [FIG. **8**A shows a K8s cluster **800***a* comprising multiple name-spaces **802-824** and multiple cores **842** and **844**. FIG. **8**B shows a K8s cluster **800***b* comprising one name-space **850** and a core **852**.

In the example of FIG. **8**A, as a non-limiting representative example, the K8s cluster **800***a* comprises multiple NFs **108** separated into a core **842** and a core **844**. In FIG. **8**A, each of the name-spaces **802-824** is mapped to the core **842** or the core **844**. In some embodiments, the name-space IDs **162** may reference one of the name-spaces **802-824** to access one NF **108** or multiple NFs **108**. The core **842** comprises a name-space **802**, a name-space **812**, and a name-space **814**. The name-space **802** may be referenced to access an AMF **848***a* and a SMF **848***b* in the core **842**, the name-space **812** may be referenced to access the AMF **848***a* in the core **842**, and the name-space **814** may be referenced to access the SMF **848***b* in the core **842**. In some embodiments, the network access commands **142** may be configured to provide access to the AMF **848***a* and/or the SMF **848***b* via the name-space **802**, the name-space **812**, and the name-space **814**. The core **844** comprises a name-space **804**, a name-space **822**, and a name-space **824**. The name-space **804** may be referenced to access an AMF **848***c*, an AMF **848***d*, a SMF **848***e*, and a SMF **848***f* in the core **844**, the name-space **822** may be referenced to access the AMF **848***c* and the AMF **848***d* in the core **844**, and the name-space **824** may be referenced to access the SMF **848***e* and the SMF **848***f* in the core **844**. In some embodiments, the network access commands **142** may be configured to provide access to the AMF **848***c*, the AMF **848***d*, the SMF **848***e*, and/or the SMF **848***f* via the name-space **804**, the name-space **822**, and the name-space **824**. Further, the network access commands **142** may be configured to provide access to the AMF **848***c* and/or the AMF **848***d* in a function set **830**.

In the example of FIG. **8**B, as a non-limiting representative example, the K8s cluster **800***b* comprises multiple NFs **108** in a single core **872**. In FIG. **8**B, the name-space **802** and the name-space **804** are mapped to the core **872**. In some embodiments, the name-space IDs **162** may reference the name-space **802** and the name-space **804** to access a portion of the NFs **108** or all the NFs **108**. The name-space **802** and the name-space **804** may be referenced to access an NRF **848***g*, a PCF **848***h*, and a UDR **848***i* in the core **872**. In some embodiments, the network access commands **142** may be configured to provide access to the NRF **848***g*, the PCF **848***h*, and/or the UDR **848***i* in the core **872** via the name-space **802** and the name-space **804**. Further, the network access commands **142** may be configured to provide access to the NRF **848***g*, the PCF **848***h*, and/or the UDR **848***i*. In this regard, the K8s cluster **800***b* may be a single cluster shared by multiple name-spaces and comprising NFs **108** associated with multiple cores. For example, the core **872**, may be a combined extension of the core **842** and the core **844** of FIG. **8**A.

In one or more embodiments, while the K8s cluster **800***a* and the K8s cluster **800***b* show certain NFs **108**, the K8s cluster **800***a* and the K8s cluster **800***b* may comprise less or more NFs **108**. The NRF **848***g* may perform one or more operations similar to those described in reference to the NRF **108***a* of FIG. **1**. The PCF **848***h* may perform one or more operations similar to those described in reference to the PCF **108***e* of FIG. **1**. The UDR **848***i* may perform one or more operations similar to those described in reference to the UDR **108***f* of FIG. **1**.

Example Process to Implement Name-Spaces in Hierarchical Multi-Tenant Containerized Service Clusters

FIG. **9** illustrates an example flowchart of a process **900** to implement name-spaces in hierarchical multi-tenant k8s clusters, in accordance with one or more embodiments. In one or more embodiments, the process **900** comprises operations **902-414**. Modifications, additions, or omissions may be made to the process **900**. The process **900** may include more, fewer, or other operations than those shown below. For example, operations may be performed in parallel or in any suitable order. While at times discussed as the server **102**, one or more of the user equipment **116**, components of any of thereof, or any suitable system or components of the communication system **100** may perform one or more operations of the process **900**. For example, one or more operations of the process **900** may be implemented, at least in part, in the form of server instructions **130** of FIG. **1**, stored on non-transitory computer readable media, tangible media, machine-readable media (e.g., server memory **128** of FIG. **1** operating as a non-transitory computer readable medium) that when run by one or more processors (e.g., the server processor **120** of FIG. **1**) may cause the one or more processors to perform operations described in operations **902-914** of the process **900**.

The process **900** starts at operation **902**, where the server **102** receives an incoming request **132** to access a network function **108** in a K8 cluster **155**. At operation **904**, the server **102** determines a tenant profile **136** and a name-space ID **162** in the incoming request. The request **132** may comprise one or more network IDs **160** associated with one or more name-space IDs **162** and/or one or more slice group IDs **164**. At operation **906**, the server **102** determine network functions **108** associated with the tenant profile **136** and the name-space ID **162**. The name-space ID **162** may indicate a name-space (e.g., name-spaces **502-524** of FIG. **5**) located in a K8s cluster **155**.

The process **900** continues at operation **910**, where the server **102** is configured to determine whether the tenant profile **136** is entitled to access the name-space. In this regard, the server **102** may determine network access commands **142** based at least in part upon the tenant profile **136** and the name-space ID **162**. The network access commands **142** may be configured to provide access to one or more entitlements **156** and enable access to the name-space in the K8s cluster **155**. If the server **102** determines that the tenant profile is not entitled to access the name-space (i.e., NO), the process **900** proceeds to operation **912**. At operation **912**, the server **102** indicates that the tenant profile is not entitled to access a name-space corresponding to the name-space ID as an alert. The alert may be a visual alert or a sound alert presented to one or more users **129** via a corresponding user equipment **116**. If the server **102** determines that the tenant profile is entitled to access the name-space (i.e., YES), the process **900** proceeds to operation **914**.

In this case, the process **900** may conclude at operation **914**, where the server **102** provides access commands indicating access to one or more K8s cluster that matches the name-space ID. The server **102** may be configured to generate a report **158** (e.g., a signal or a communication) indicating or comprising information indicating the network access commands **142**. In this regard, the server **102** may present the report **158** to a user equipment **116** configured to access the service **106** based at least in part upon the network access commands **142** in the report **158**.

Scope of the Disclosure

While several embodiments have been provided in the present disclosure, it should be understood that the disclosed systems and methods might be embodied in many other specific forms without departing from the spirit or scope of the present disclosure. The present examples are to be considered as illustrative and not restrictive, and the intention is not to be limited to the details given herein. For example, the various elements or components may be combined or integrated with another system or certain features may be omitted, or not implemented.

In addition, techniques, systems, subsystems, and methods described and illustrated in the various embodiments as discrete or separate may be combined or integrated with other systems, modules, techniques, or methods without departing from the scope of the present disclosure. Other items shown or discussed as coupled or directly coupled or communicating with each other may be indirectly coupled or communicating through some interface, device, or intermediate component whether electrically, mechanically, or otherwise. Other examples of changes, substitutions, and alterations are ascertainable by one skilled in the art and could be made without departing from the spirit and scope disclosed herein.

To aid the Patent Office, and any readers of any patent issued on this application in interpreting the claims appended hereto, applicants note that they do not intend any of the appended claims to invoke 35 U.S.C. § 112(f) as it exists on the date of filing hereof unless the words "means for" or "step for" are explicitly used in the particular claim.

The invention claimed is:

1. An apparatus, comprising:
- a memory, comprising:
  - one or more directories comprising access to a plurality of tenant profiles, each tenant profile of the plurality of tenant profiles being associated with one or more services of a plurality of services; and
  - one or more network access commands configured to provide access to one or more entitlements; and
- a processor communicatively coupled to the memory and configured to:
  - receive a first request to access at least one service of the plurality of services, the first request comprising a first application function identifier (AFID);
  - extrapolate a first tenant identifier (ID), a first department ID, and a first application programming interface (API) ID from the first AFID, wherein:
    - the first tenant ID references a first tenant profile of the plurality of tenant profiles;
    - the first department ID references a first plurality of entitlements associated with the first tenant profile; and
    - the first API ID references a first service associated with the first plurality of entitlements;
  - determine a first plurality of network access commands configured to enable access to the first service in accordance with the first plurality of entitlements; and
  - generate a first report comprising the first plurality of network access commands.

2. The apparatus of claim **1**, wherein the AFID comprises a first plurality of characters corresponding to the tenant ID, a second plurality of characters corresponding to the first department ID, and a third plurality of characters corresponding to the API ID.

3. The apparatus of claim **1**, wherein the AFID is an information element that comprises an availability between 50 characters and 150 characters.

4. The apparatus of claim **1**, wherein the processor is further configured to:
- receive a second request comprising a second AFID;
- extrapolate a second tenant ID, a second department ID, and a second API ID from the second AFID, wherein:
  - the second tenant ID references a second tenant profile of the plurality of tenant profiles;
  - the second department ID references a second plurality of entitlements associated with the second tenant profile; and
  - the second API ID references a second service and a third service associated with the second plurality of entitlements;
- determine a second plurality of network access commands configured to enable access to the second service in accordance with the second plurality of entitlements and a third plurality of network access commands configured to enable access to the third service in accordance with the second plurality of entitlements; and
- generate a second report comprising the second plurality of network access commands and the third plurality of network access commands.

5. The apparatus of claim **1**, wherein the processor is further configured to:
- receive a second request comprising a second AFID;
- extrapolate a second tenant ID, a second department ID, and a second API ID from the second AFID, wherein:
  - the second tenant ID references a second tenant profile of the plurality of tenant profiles;
  - the second department ID references a second plurality of entitlements and a third plurality of entitlements associated with the second tenant profile; and
  - the second API ID references a second service associated with the second plurality of entitlements and the third plurality of entitlements;
- determine a second plurality of network access commands configured to enable access to the second service in accordance with the second plurality of entitlements and the third plurality of entitlements; and
- generate a second report comprising the second plurality of network access commands.

6. The apparatus of claim **1**, wherein the processor is further configured to:
- receive a second request comprising a second AFID;
- extrapolate a second tenant ID and a second API ID from the second AFID, wherein:
  - the second tenant ID references a second tenant profile of the plurality of tenant profiles; and
  - the second API ID references a second service associated with the second tenant ID;
- determine a second plurality of network access commands configured to enable access to the second service in accordance with a second plurality of entitlements corresponding to the second tenant profile; and
- generate a second report comprising the second plurality of network access commands.

7. The apparatus of claim **1**, wherein the processor is further configured to present the first report to a user equipment, the user equipment being configured to access the first service based at least in part upon the first plurality of network access commands in the first report.

8. A method, comprising:

receiving a first request to access at least one service of a plurality of services, the first request comprising a first application function identifier (AFID);

extrapolating a first tenant identifier (ID), a first department ID, and a first application programming interface (API) ID from the first AFID, wherein:

the first tenant ID references a first tenant profile of the plurality of tenant profiles;

the first department ID references a first plurality of entitlements associated with the first tenant profile; and

the first API ID references a first service associated with the first plurality of entitlements;

determining a first plurality of network access commands configured to provide access to one or more entitlements and enable access to the first service in accordance with the first plurality of entitlements; and

generating a first report comprising the first plurality of network access commands.

9. The method of claim **8**, wherein the AFID comprises a first plurality of characters corresponding to the tenant ID, a second plurality of characters corresponding to the first department ID, and a third plurality of characters corresponding to the API ID.

10. The method of claim **8**, wherein the AFID is an information element that comprises an availability between 50 characters and 150 characters.

11. The method of claim **8**, further comprising:

receiving a second request comprising a second AFID;

extrapolating a second tenant ID, a second department ID, and a second API ID from the second AFID, wherein:

the second tenant ID references a second tenant profile of the plurality of tenant profiles;

the second department ID references a second plurality of entitlements associated with the second tenant profile; and

the second API ID references a second service and a third service associated with the second plurality of entitlements;

determining a second plurality of network access commands configured to enable access to the second service in accordance with the second plurality of entitlements and a third plurality of network access commands configured to enable access to the third service in accordance with the second plurality of entitlements; and

generating a second report comprising the second plurality of network access commands and the third plurality of network access commands.

12. The method of claim **8**, further comprising:

receiving a second request comprising a second AFID;

extrapolating a second tenant ID, a second department ID, and a second API ID from the second AFID, wherein:

the second tenant ID references a second tenant profile of the plurality of tenant profiles;

the second department ID references a second plurality of entitlements and a third plurality of entitlements associated with the second tenant profile; and

the second API ID references a second service associated with the second plurality of entitlements and the third plurality of entitlements;

determining a second plurality of network access commands configured to enable access to the second service in accordance with the second plurality of entitlements and the third plurality of entitlements; and

generating a second report comprising the second plurality of network access commands.

13. The method of claim **8**, further comprising:

receiving a second request comprising a second AFID;

extrapolating a second tenant ID and a second API ID from the second AFID, wherein:

the second tenant ID references a second tenant profile of the plurality of tenant profiles; and

the second API ID references a second service associated with the second tenant ID;

determining a second plurality of network access commands configured to enable access to the second service in accordance with a second plurality of entitlements corresponding to the second tenant profile; and

generating a second report comprising the second plurality of network access commands.

14. The method of claim **8**, further comprising presenting the first report to a user equipment, the user equipment being configured to access the first service based at least in part upon the first plurality of network access commands in the first report.

15. A non-transitory computer readable medium storing instructions that when executed by a processor cause the processor to:

receive a first request to access at least one service of a plurality of services, the first request comprising a first application function identifier (AFID);

extrapolate a first tenant identifier (ID), a first department ID, and a first application programming interface (API) ID from the first AFID, wherein:

the first tenant ID references a first tenant profile of the plurality of tenant profiles;

the first department ID references a first plurality of entitlements associated with the first tenant profile; and

the first API ID references a first service associated with the first plurality of entitlements;

determine a first plurality of network access commands configured to provide access to one or more entitlements and enable access to the first service in accordance with the first plurality of entitlements; and

generate a first report comprising the first plurality of network access commands.

16. The non-transitory computer readable medium of claim **15**, wherein the AFID comprises a first plurality of characters corresponding to the tenant ID, a second plurality of characters corresponding to the department ID, and a third plurality of characters corresponding to the API ID.

17. The non-transitory computer readable medium of claim **15**, wherein the AFID is an information element that comprises an availability between 50 characters and 150 characters.

18. The non-transitory computer readable medium of claim **15**, wherein the processor is further caused to:

receive a second request comprising a second AFID;

extrapolate a second tenant ID, a second department ID, and a second API ID from the second AFID, wherein:

the second tenant ID references a second tenant profile of the plurality of tenant profiles;

the second department ID references a second plurality of entitlements associated with the second tenant profile; and

the second API ID references a second service and a third service associated with the second plurality of entitlements;

determine a second plurality of network access commands configured to enable access to the second service in accordance with the second plurality of entitlements and a third plurality of network access commands configured to enable access to the third service in accordance with the second plurality of entitlements; and

generate a second report comprising the second plurality of network access commands and the third plurality of network access commands.

19. The non-transitory computer readable medium of claim **15**, wherein the processor is further caused to:

receive a second request comprising a second AFID;

extrapolate a second tenant ID, a second department ID, and a second API ID from the second AFID, wherein:

the second tenant ID references a second tenant profile of the plurality of tenant profiles;

the second department ID references a second plurality of entitlements and a third plurality of entitlements associated with the second tenant profile; and

the second API ID references a second service associated with the second plurality of entitlements and the third plurality of entitlements;

determine a second plurality of network access commands configured to enable access to the second service in accordance with the second plurality of entitlements and the third plurality of entitlements; and

generate a second report comprising the second plurality of network access commands.

20. The non-transitory computer readable medium of claim **15**, wherein the processor is further caused to present the first report to a user equipment, the user equipment being configured to access the first service based at least in part upon the first plurality of network access commands in the first report.

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